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(54) **TEMPERATURE-CONTROLLED STORAGE DEVICE**

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(52) **U.S. Cl.**
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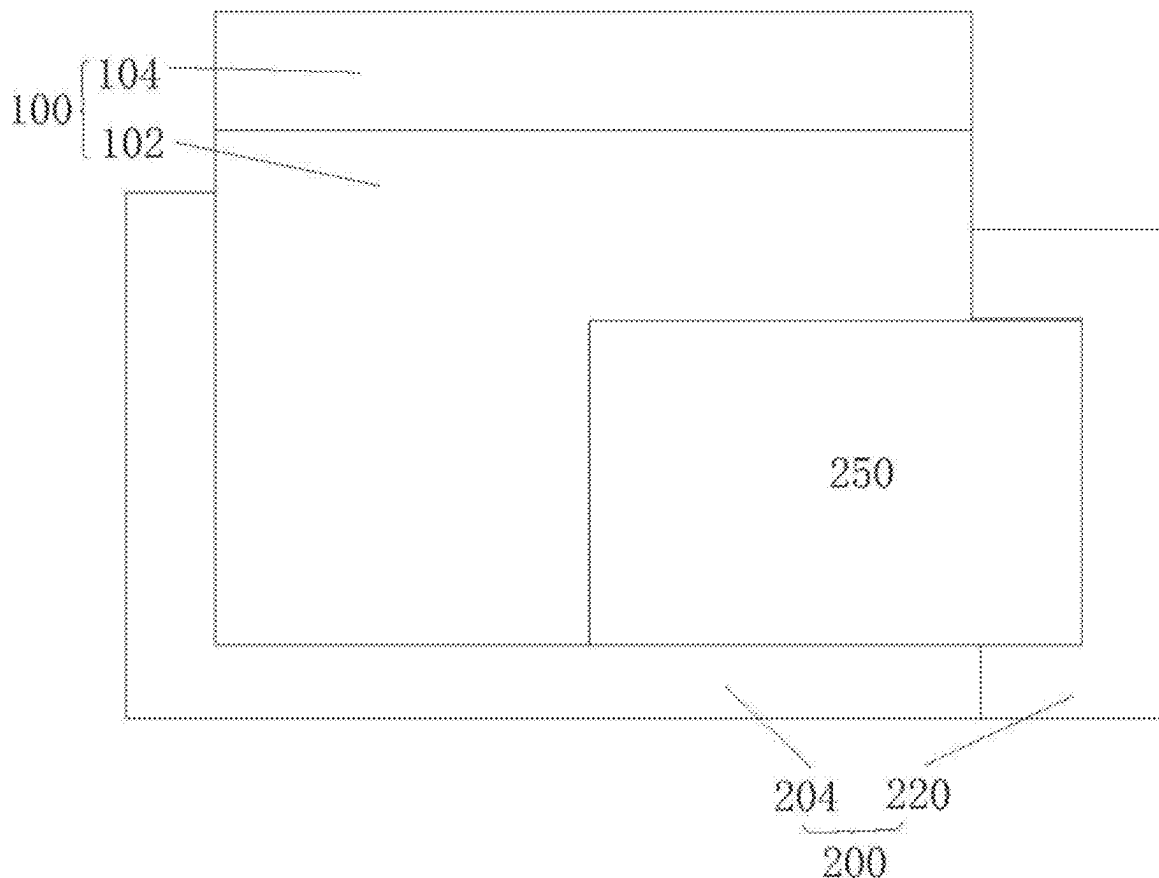
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Aug. 19, 2024	(CN)	202422005455.4
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(57) **ABSTRACT**

Disclosed is a temperature-controlled storage device, including a storage module, a soft outer package, and a temperature control module, and the soft outer package covers a periphery of the storage module; and when being powered, the temperature control module is configured for cooling, such that a temperature inside the storage module can be adjusted. By providing the soft outer package, the temperature-controlled storage device in the present disclosure has the following advantageous effects: first, the heat preservation effect of the storage device is improved; second, the soft outer package can modify an overall appearance of the storage device without increasing a weight of the device, giving the user a sense of visual and actual lightness; third, a soft appearance can improve the comfort of carrying the device; and fourth, the soft outer package can protect the temperature control module.



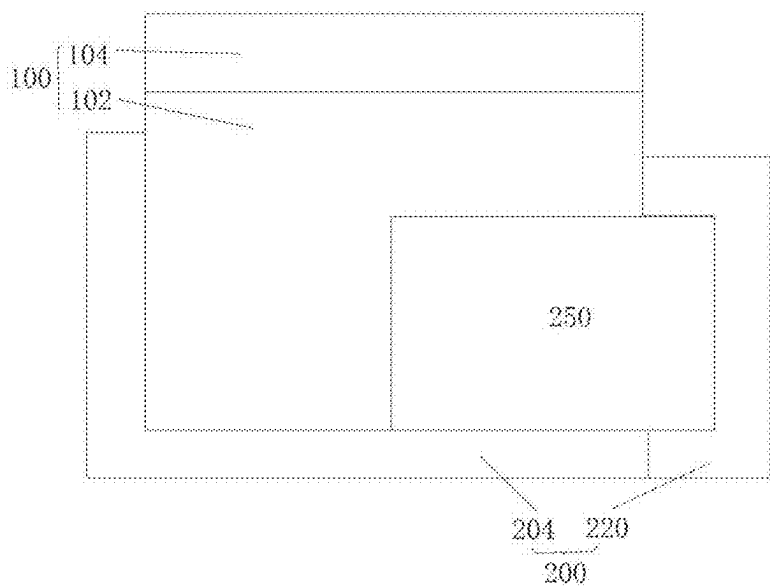


FIG. 1

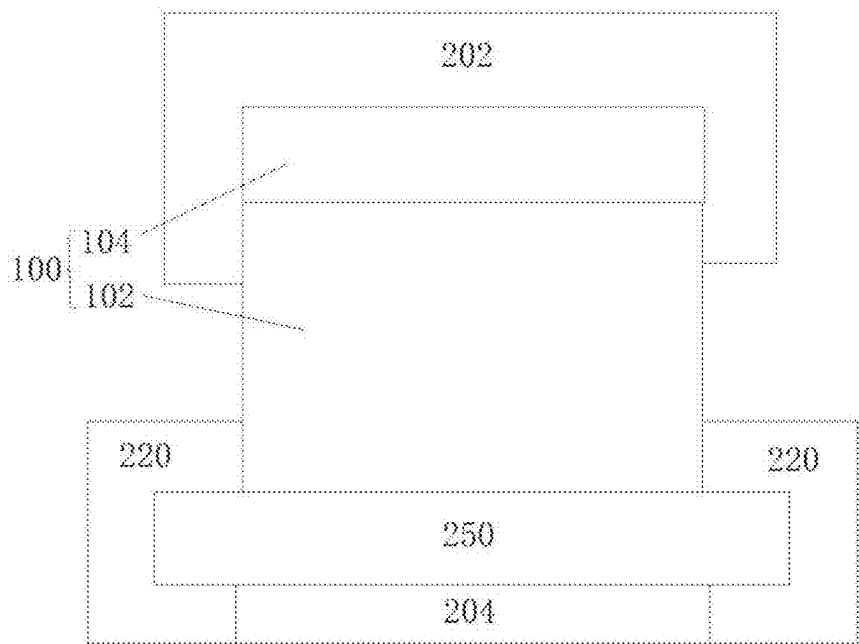


FIG. 2

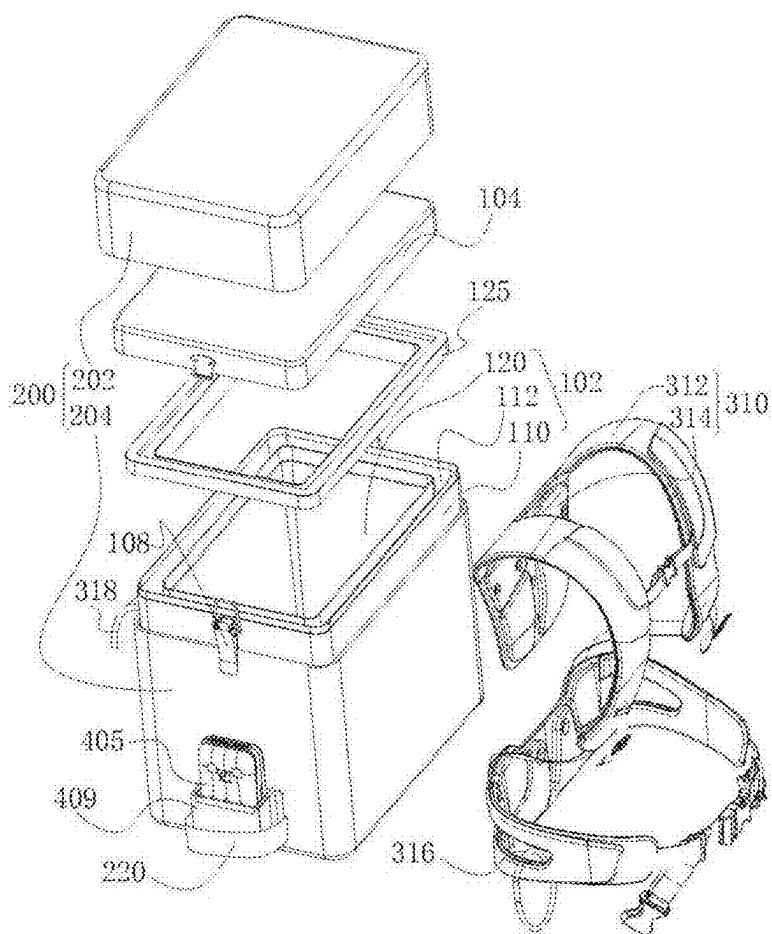


FIG. 3

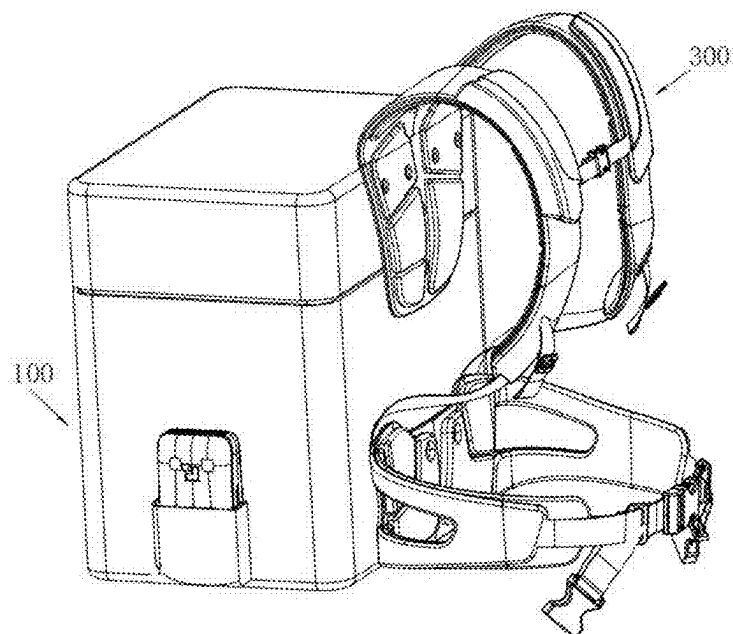


FIG. 4

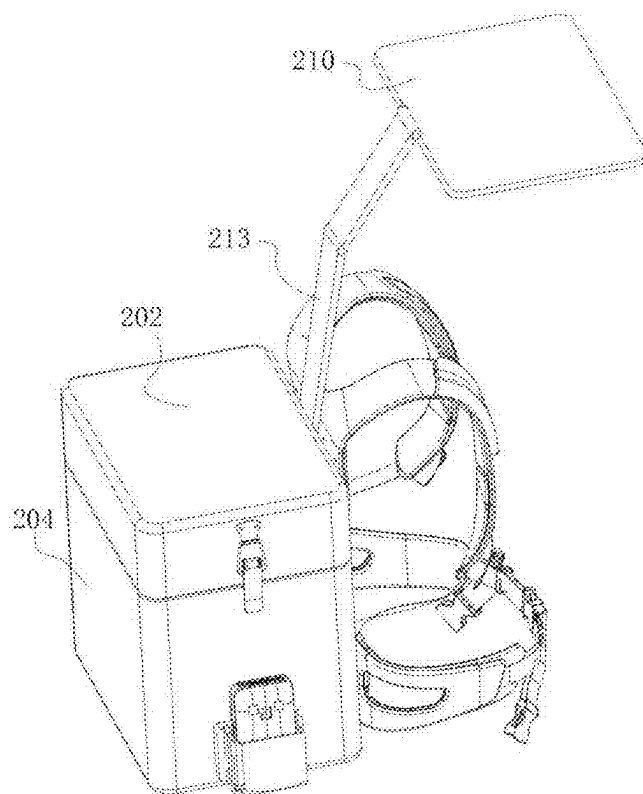


FIG. 5

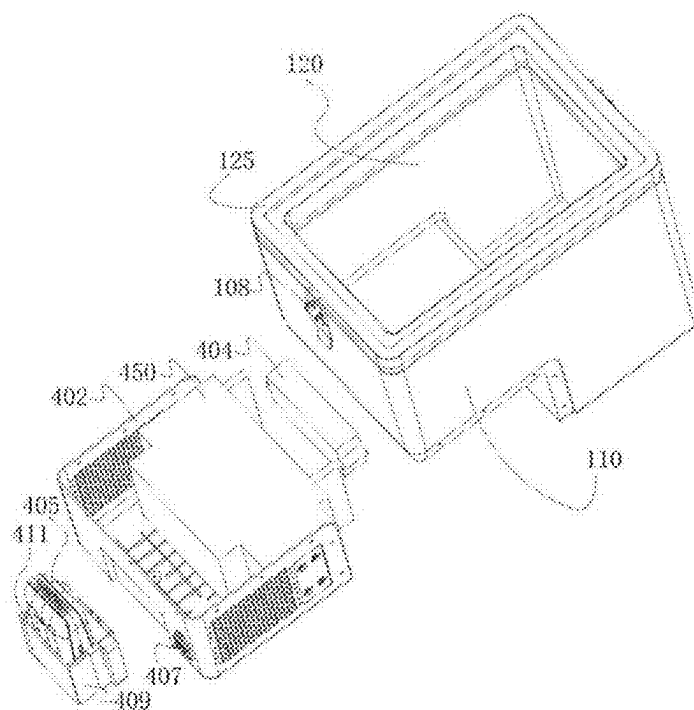


FIG. 6

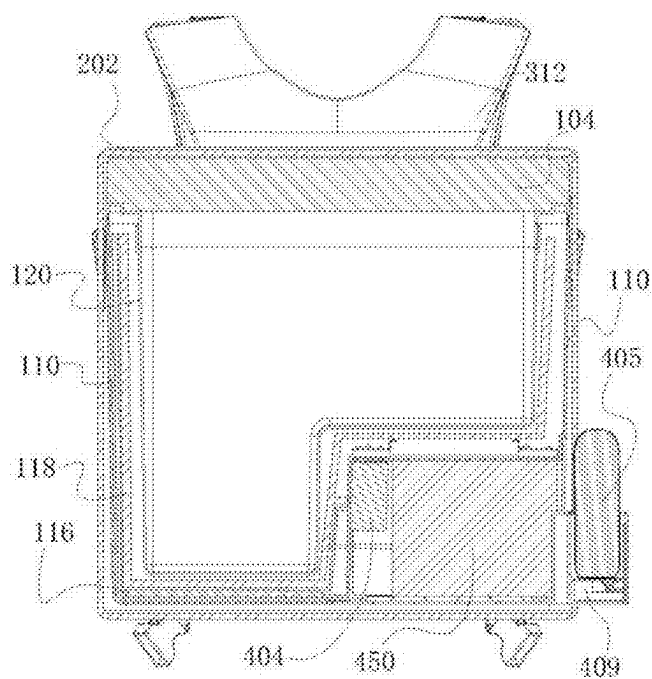


FIG. 7

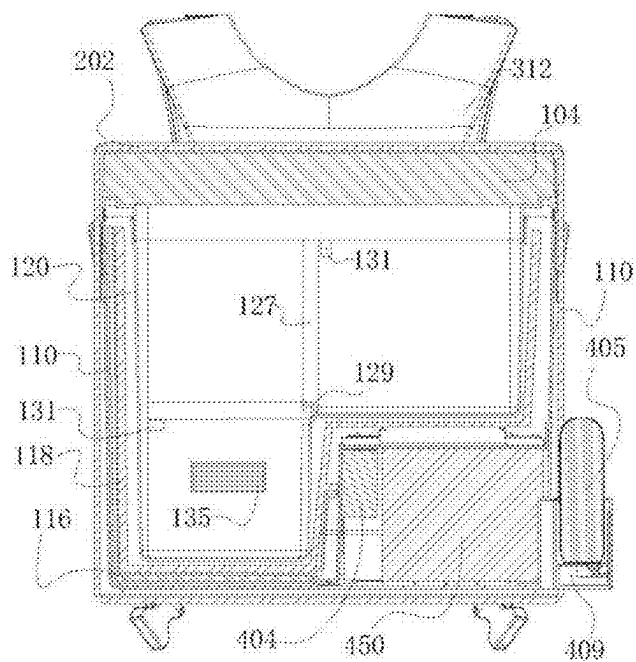


FIG. 8

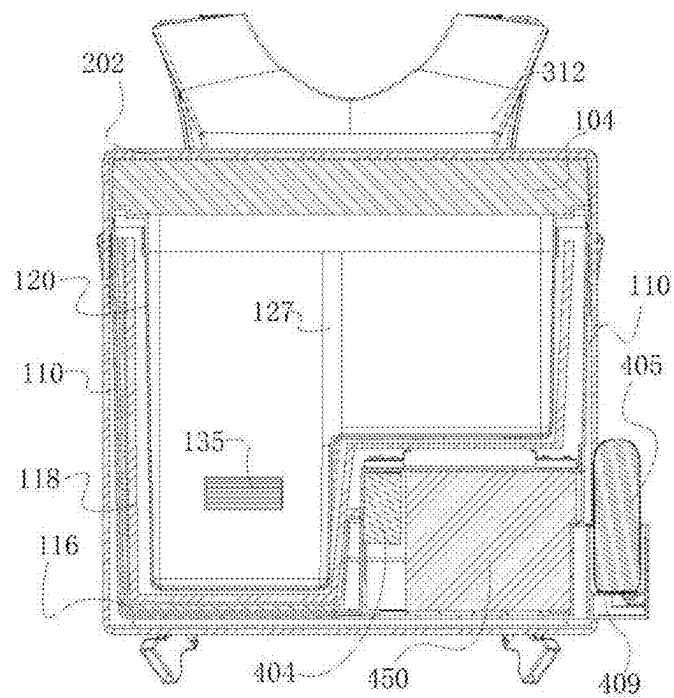


FIG. 9

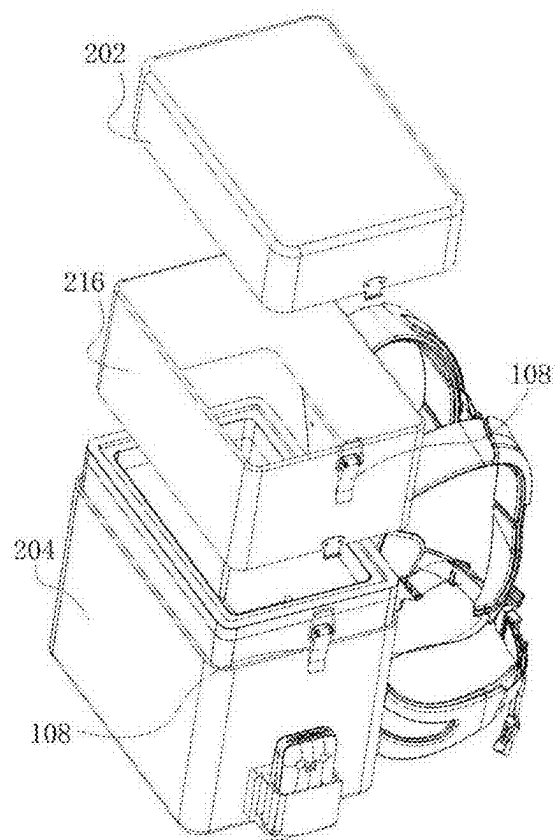


FIG. 10

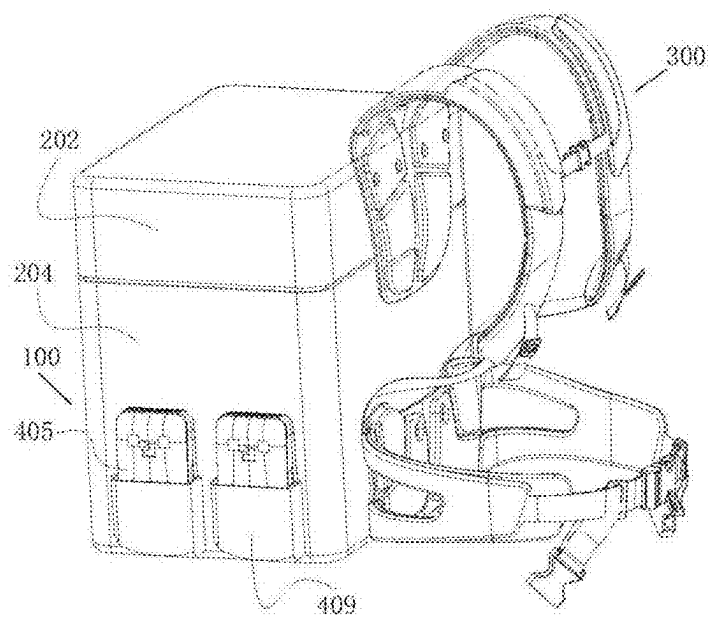


FIG. 11

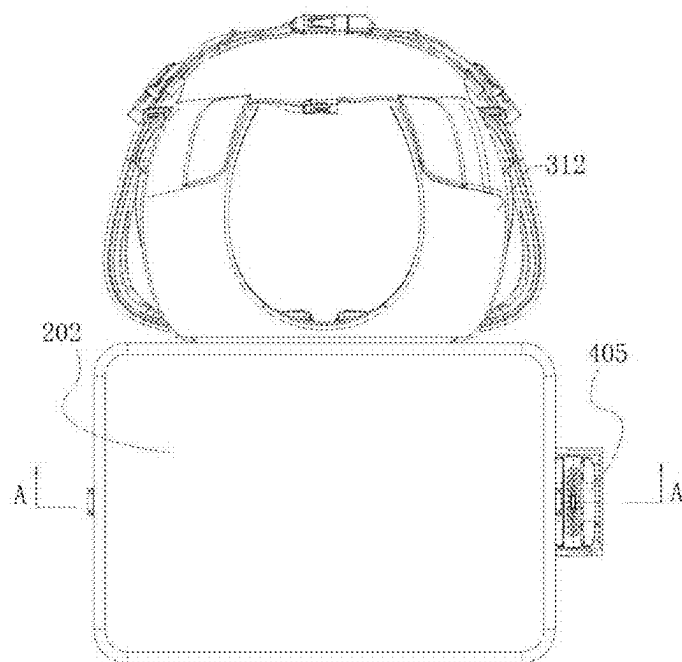


FIG. 12

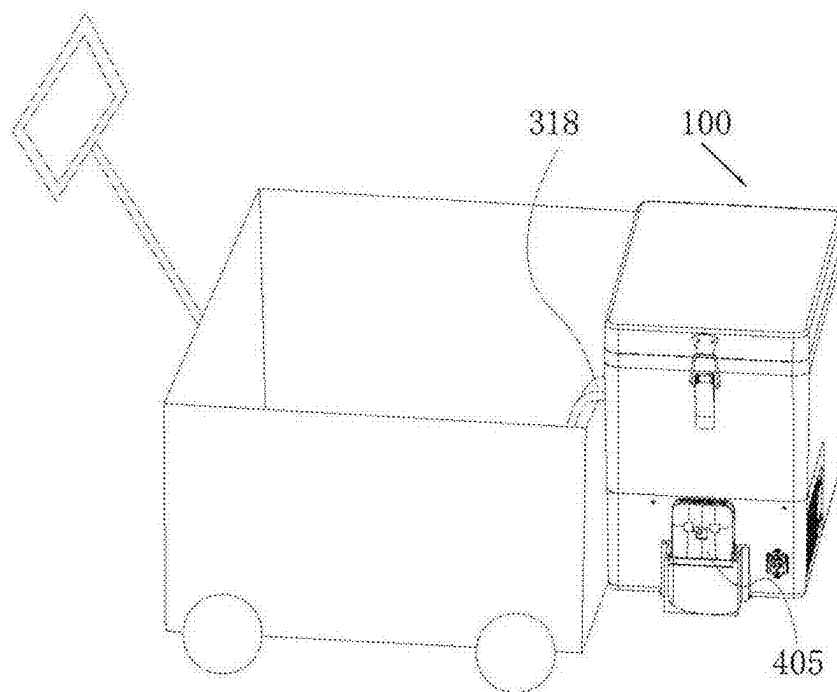


FIG. 13

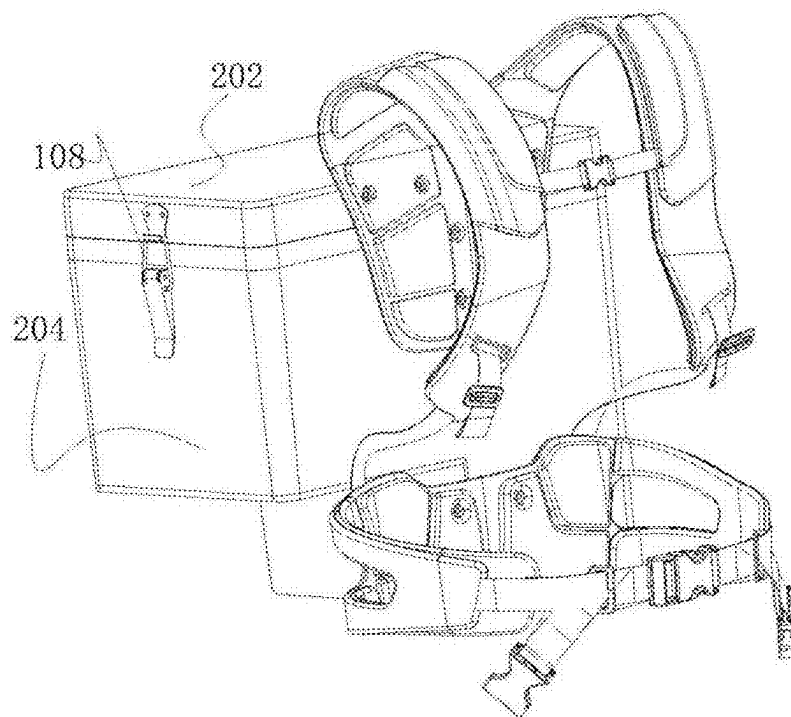


FIG. 14A

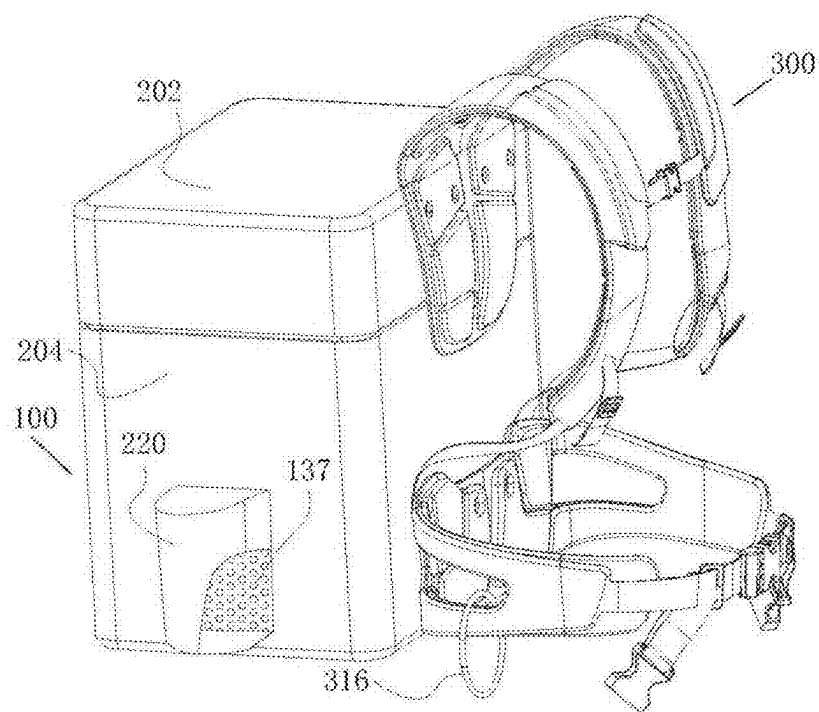


FIG. 14B

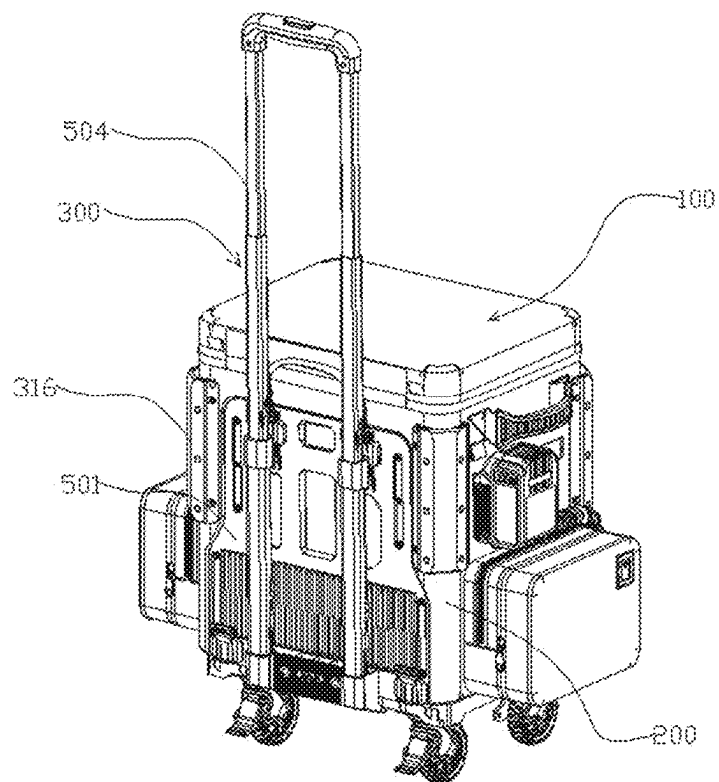


FIG. 15

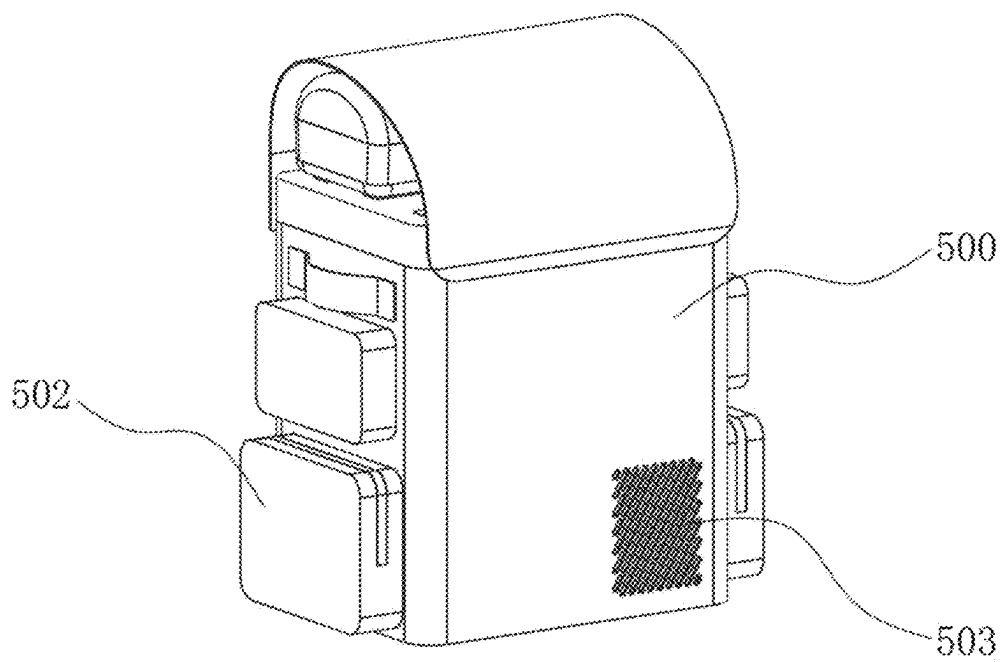


FIG. 16

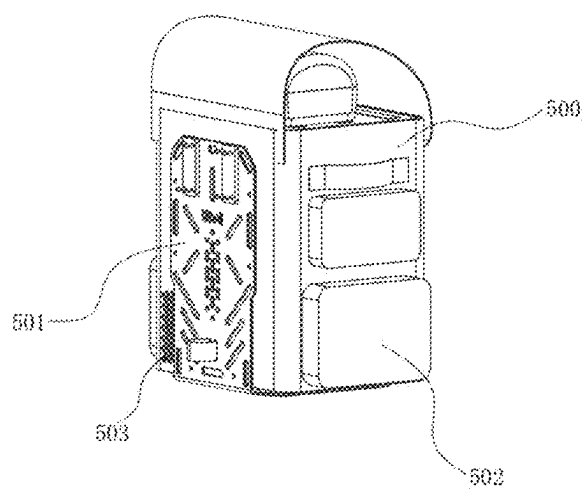


FIG. 17

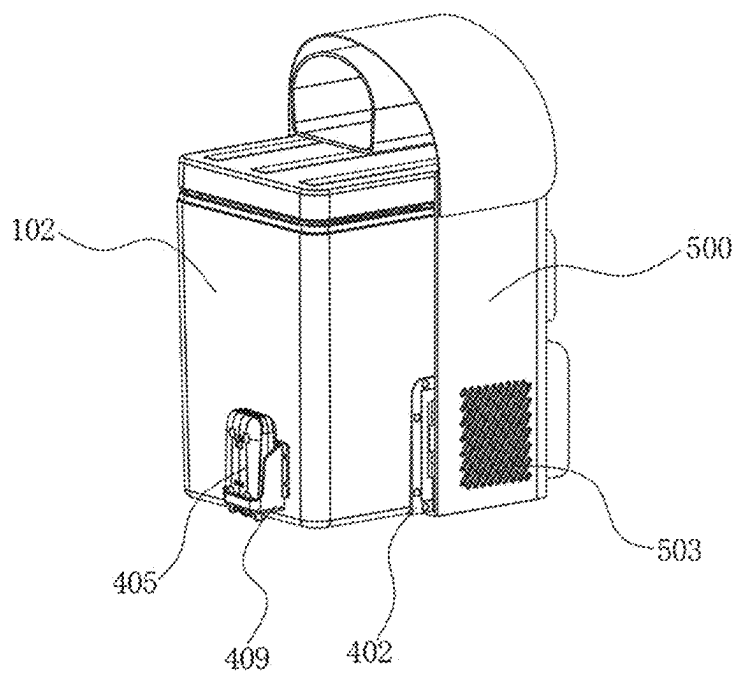


FIG. 18

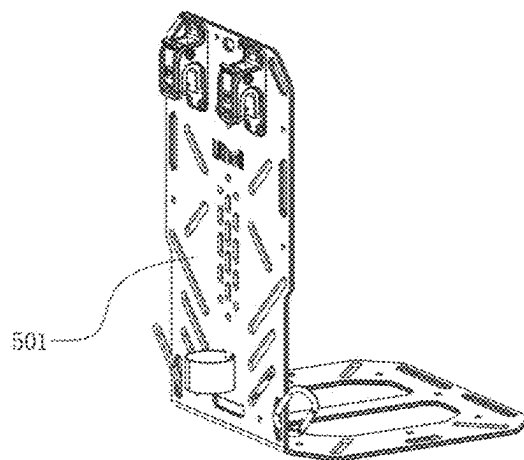


FIG. 19

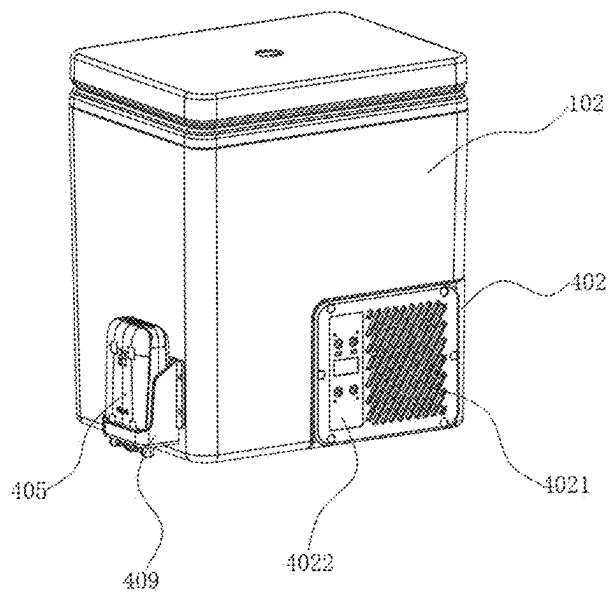


FIG. 20A

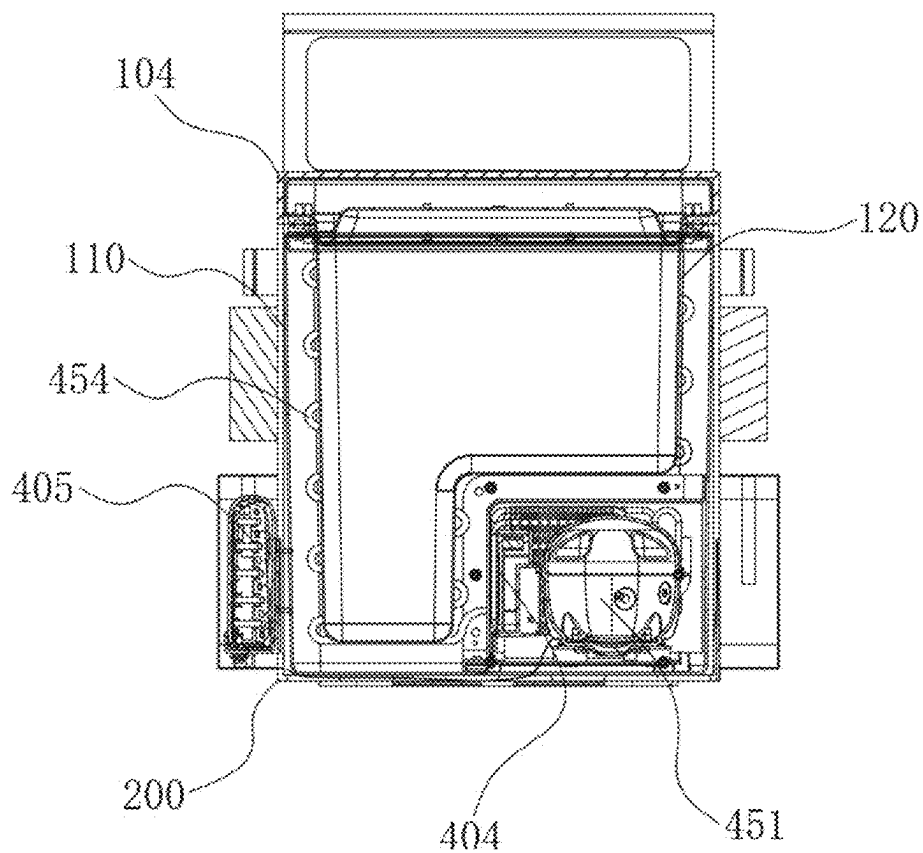


FIG. 20B

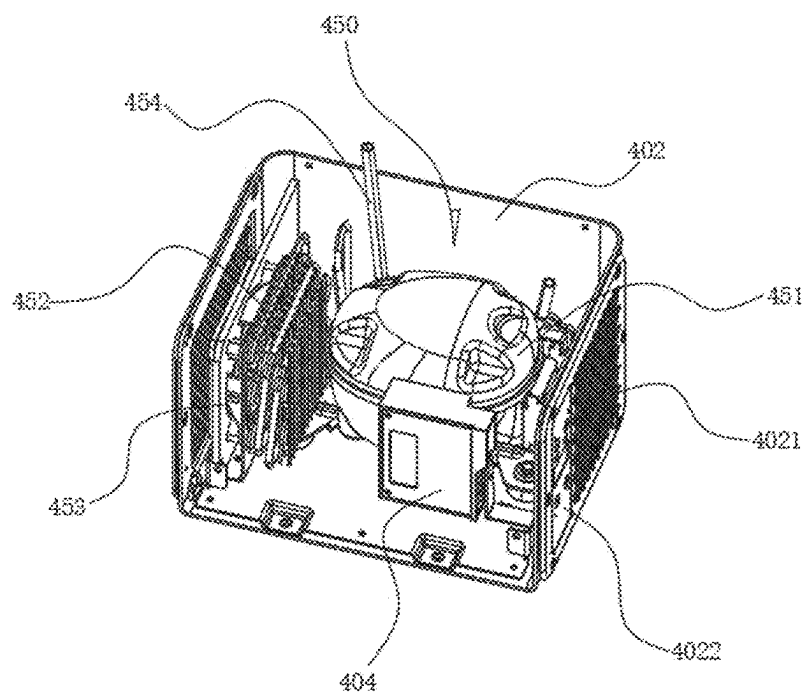


FIG. 21

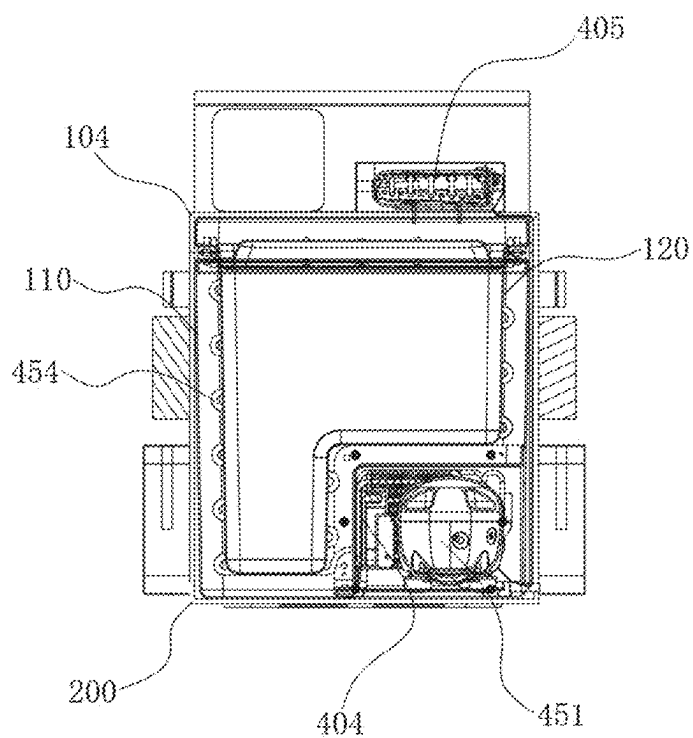


FIG. 22

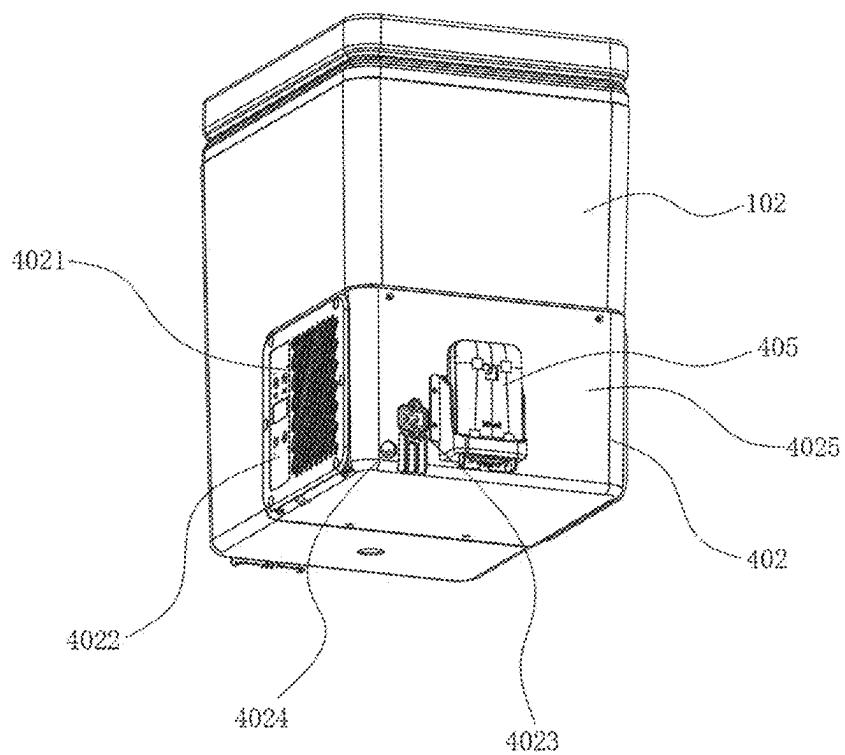


FIG. 23A

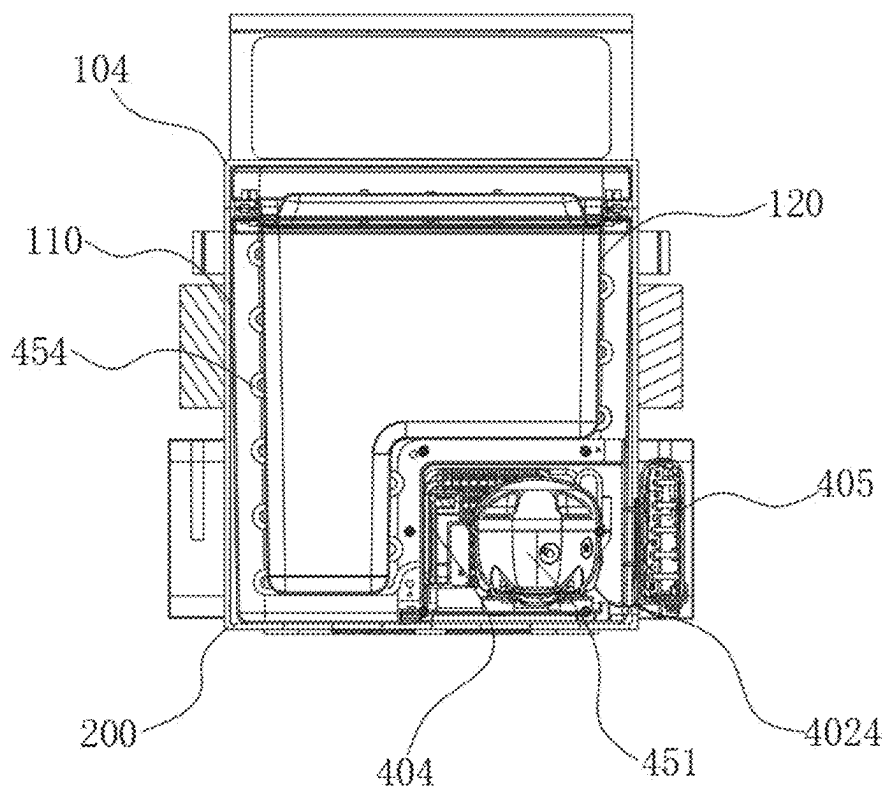


FIG. 23B

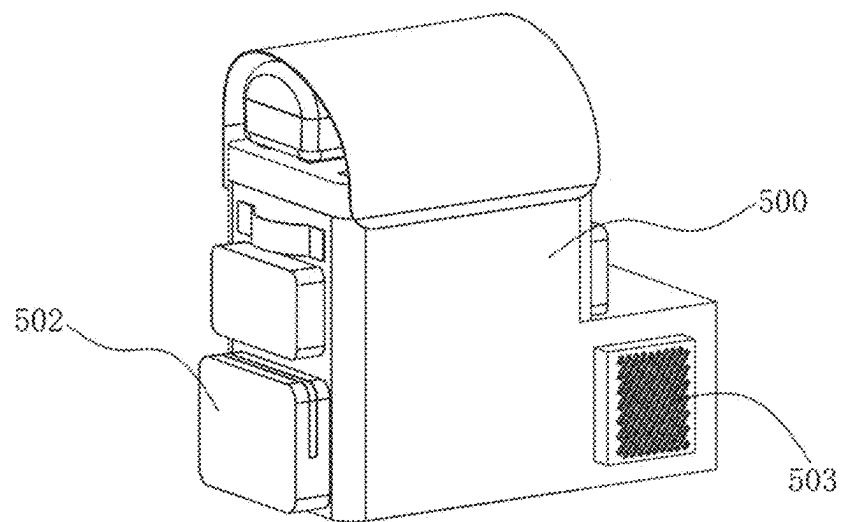


FIG. 24

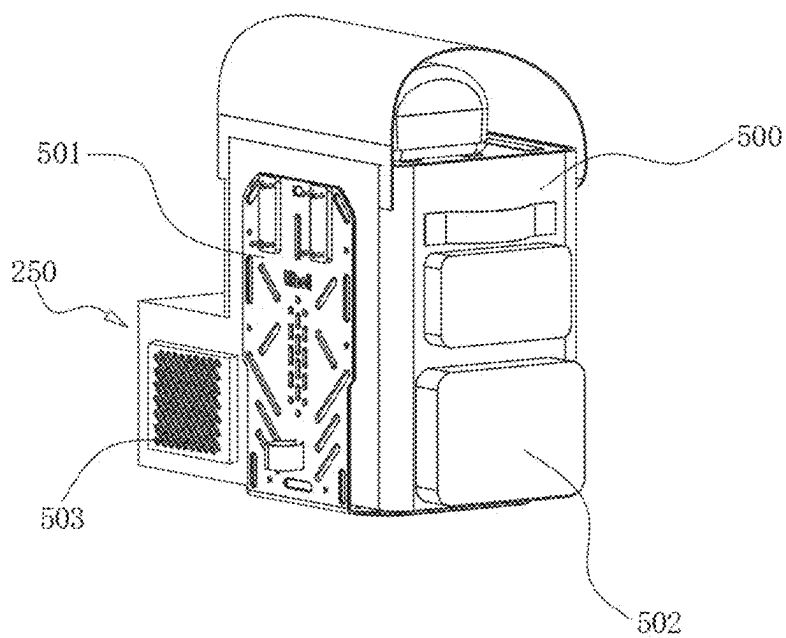


FIG. 25

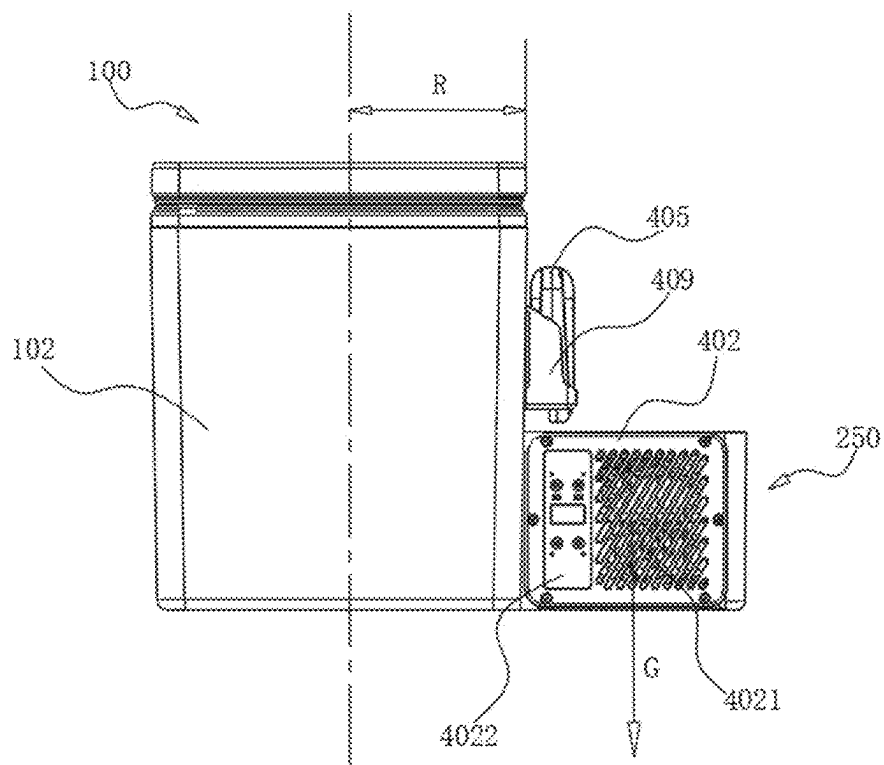


FIG. 26

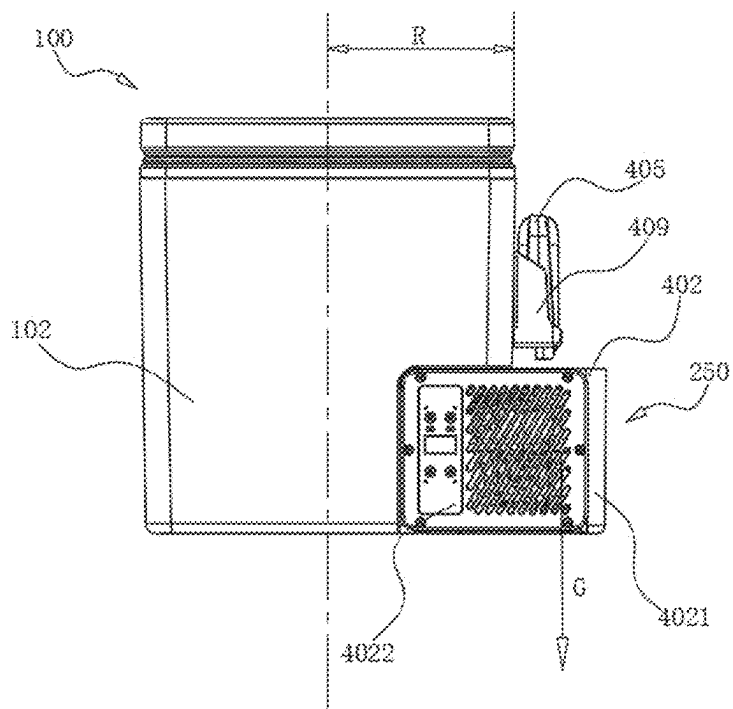


FIG. 27

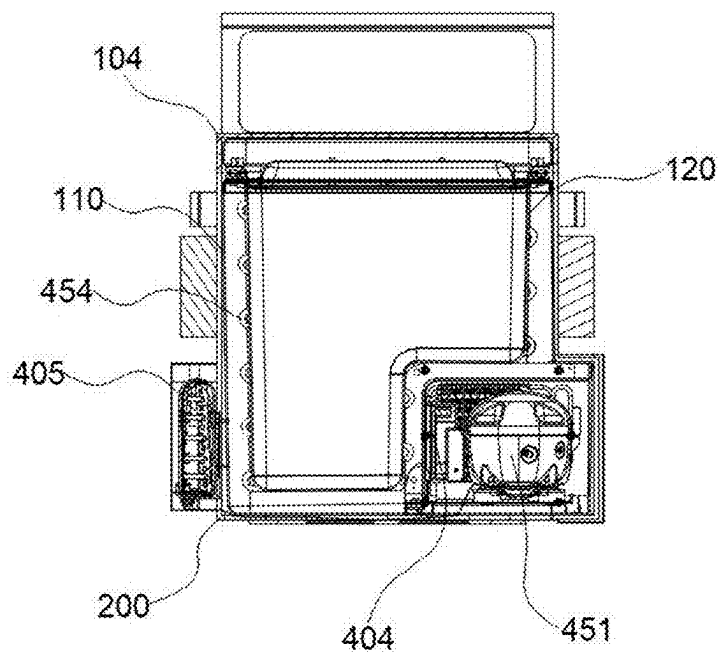


FIG. 28

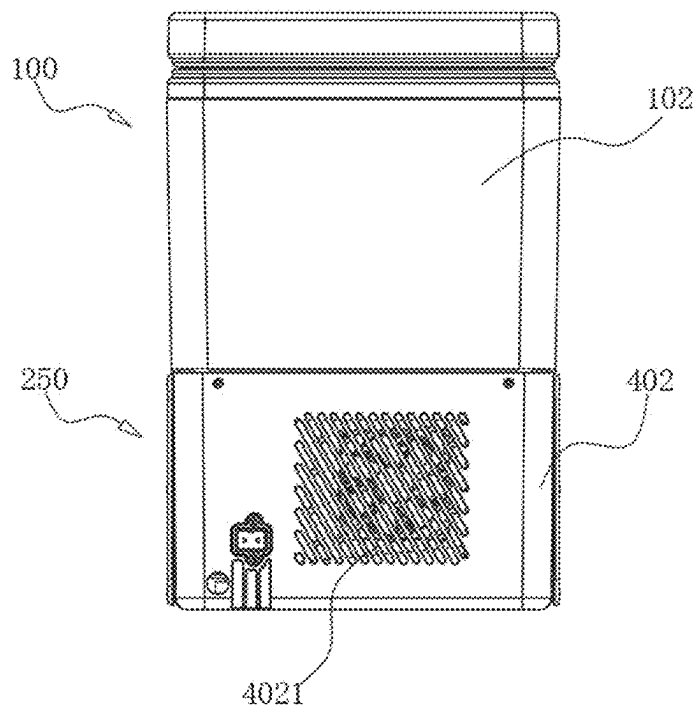


FIG. 29

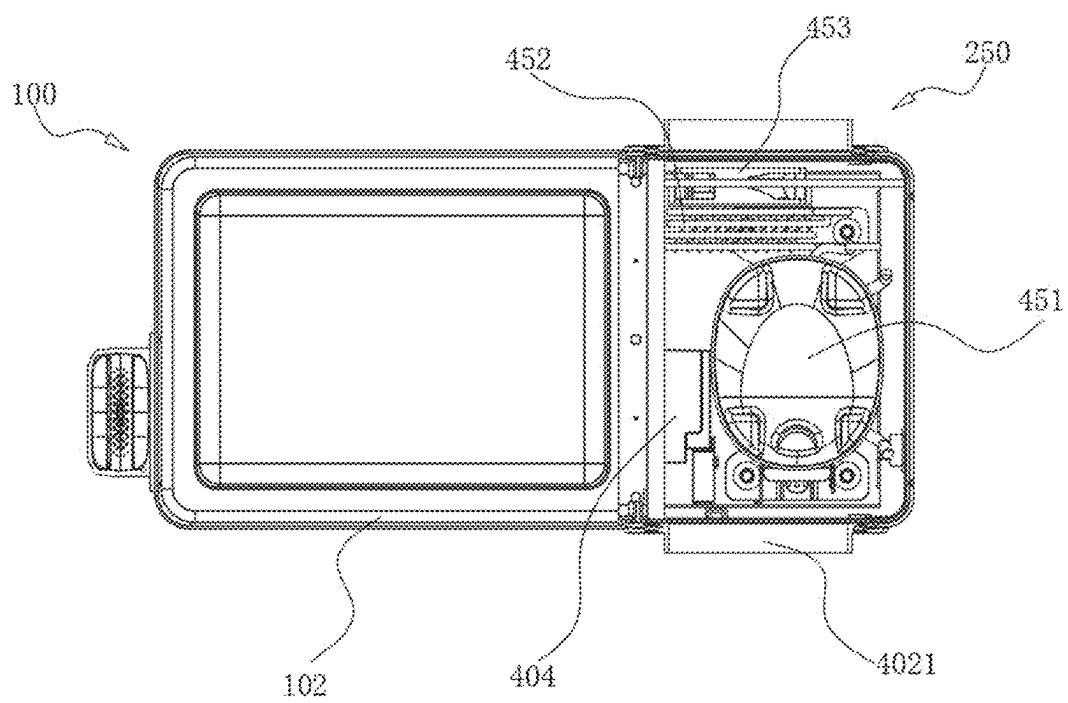


FIG. 30

TEMPERATURE-CONTROLLED STORAGE DEVICE

CROSS REFERENCE TO THE RELATED APPLICATIONS

[0001] This application is based upon and claims priority to Chinese Patent Application No. 202410216388.2 filed on Feb. 27, 2024, Chinese Patent Application No. 202410237682.1 filed on Mar. 1, 2024, Chinese Patent Application No. 202421373502.4 filed on Jun. 17, 2024, Chinese Patent Application No. 202422005455.4 filed on Aug. 19, 2024, and Chinese Patent Application No. 202422748219.1 filed on Nov. 11, 2024. The entire contents of Chinese Patent Application No. 202410216388.2, Chinese Patent Application No. 202410237682.1, Chinese Patent Application No. 202421373502.4, Chinese Patent Application No. 202422005455.4, and Chinese Patent Application No. 202422748219.1 are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of outdoor food storage, and particularly relates to a temperature-controlled storage device.

BACKGROUND

[0003] A refrigerator can offer a variety of functions, such as refrigeration, food preservation, and cooling, but it usually takes up a large space, consumes lots of electricity, and is inconvenient for outdoor use. With the rise of various outdoor recreational activities such as walking and cycling, higher requirements are raised for the storage and refrigeration of food or beverages.

[0004] In order to meet the needs for outdoor refrigeration of food or beverages, it is common practice to carry a simple insulated box, such as aluminum foil insulated box. However, the insulated box does not have good performance, only supports short-term insulation, and cannot actively adjust a temperature inside the box, thereby failing to meet the needs of refrigerating of food or beverages outdoors for an extended period of time. In addition, existing insulated boxes are inconvenient to carry, especially during walking or cycling, it is difficult for a user to carry the box for an extended period of time, which limits a range of use for existing insulated boxes.

[0005] A new temperature-controlled storage device is accordingly provided, which is convenient for the user to refrigerate food or beverages outdoors for an extended period of time.

SUMMARY

[0006] One objective of the present disclosure is to provide a temperature-controlled storage device that is conducive to miniaturization.

[0007] Another objective of the present disclosure is to provide a temperature-controlled storage device that is easy to carry.

[0008] In order to achieve at least one of the above objectives, a temperature-controlled storage device is provided according to one aspect of the present disclosure, the temperature-controlled storage device includes a storage module and a temperature control module; when being powered, the temperature control module is configured for cooling to adjust a temperature inside the storage module;

and the temperature-controlled storage device further includes a soft outer package, the soft outer package wraps a periphery of the storage module, and at least a portion of a surface of the temperature control module is wrapped by the soft outer package.

[0009] By providing the soft outer package, the temperature-controlled storage device has the following advantageous effects: first, the heat preservation effect of the storage device is improved; second, the soft outer package can modify an overall appearance of the storage device without increasing too much weight to the device, giving the user a sense of visual and actual lightness; third, a soft appearance can improve the comfort of carrying the device; and fourth, the soft outer package can protect both the storage module and the temperature control module.

[0010] In some embodiments, the temperature-controlled storage device further includes a power supply assembly, where the power supply assembly includes a battery mounting portion electrically connected to the temperature control module; a portable power source is configured for being detachably mounted in the battery mounting portion to supply power to the temperature control module; and the battery mounting portion is arranged on an outer side of the soft outer package, or the soft outer package partially wraps the battery mounting portion, or the soft outer package wraps the battery mounting portion in an openable and closable manner.

[0011] In some embodiments, at least a portion of the temperature control module protrudes from a radial perimeter surface of the storage module, and/or at least a portion of the battery mounting portion also protrudes from the radial perimeter surface of the storage module.

[0012] By protruding the temperature control module and/or battery mounting portion and covering the soft outer package, the temperature-controlled storage device has the following advantageous effects: more space can be left inside the storage module for storage purposes, thereby increasing a storage volume while maintaining the compactness of the storage module; further, the soft outer package can also modify the irregular shape and make the overall appearance of the storage is more visually coordinated.

[0013] In some embodiments, at least a portion of the temperature control module protrudes from the radial perimeter surface of the storage module, at least a portion of the battery mounting portion also protrudes from the radial perimeter surface of the storage module, and portions of the temperature control module and the battery mounting portion that protrude from the radial perimeter surface of the storage module are symmetrically located on both sides of the storage body.

[0014] In some embodiments, the storage module includes a storage body and a storage lid which is disposed on the storage body in an openable and closable manner; and the temperature control module includes a refrigeration base and a refrigeration assembly arranged inside the refrigeration base, the refrigeration base is connected to the storage body, and the battery mounting portion is electrically connected to the refrigeration assembly.

[0015] In some embodiments, a bottom of the storage body is stepped or only one corner of the bottom of the storage body is recessed inward, such that a concave space is formed at the bottom of the storage body; the temperature control module is detachably arranged in the concave space at the bottom of the storage body, and the temperature

control module does not protrude from a radial perimeter surface of the storage body; the battery mounting portion is connected to the refrigeration base, and the battery mounting portion protrudes from the radial perimeter surface of the storage body; or the battery mounting portion is connected to the radial perimeter surface of the storage body.

[0016] In some embodiments, a bottom of the storage body is stepped or only one corner of the bottom of the storage body is recessed inward, such that a concave space is formed at the bottom of the storage body; the temperature control module is detachably arranged in the concave space at the bottom of the storage body, and at least one side of the temperature control module protrudes from the radial perimeter surface of the storage body; the battery mounting portion is connected to the storage body, the battery mounting portion is located on a side away from the temperature control module, or the battery mounting portion and the temperature control module are located on a same side; or the battery mounting portion is connected to the storage lid.

[0017] In some embodiments, a center of gravity of the temperature control module is far away from the storage module, and the center of gravity (G) of the temperature control module is located in a range of 1-1.5 times a radius (R) of a central axis of the storage body.

[0018] In some embodiments, the soft outer package includes a soft pack body, the soft pack body wraps the storage body, the soft pack body wraps at least a portion of the surface of the temperature control module, and the soft pack body is assembled integrally with the storage body.

[0019] In some embodiments, the soft outer package includes a soft pack cover, the soft pack cover wraps the storage lid, and the soft pack cover is assembled integrally with the storage lid.

[0020] Further, the soft outer package also includes at least one ear pouch, the ear pouch partially wraps the battery mounting portion, or the ear pouch wraps the battery mounting portion in an openable and closable manner.

[0021] In some embodiments, the storage body and the storage lid define a sealing connection position, while the soft pack body and the soft pack cover define an accommodating connection position, and the sealing connection position and the accommodating connection position are arranged in a staggered manner in a height direction; or the storage body and the storage lid define a sealing connection position, a top of the soft pack body extends upward beyond the sealing connection position, or a bottom of the soft outer cover extends downward beyond the sealing connection position; or the storage body and the storage lid define a sealing connection position, while the soft pack body and the soft pack cover define an accommodating connection position, and the sealing connection position and the accommodating connection position are at a same level.

[0022] In some embodiments, the storage body is provided with a partition plate, the partition plate divides a storage space of the storage body into a first zone and a second zone, an air outlet for outputting cold air is formed in one of the first zone and the second zone, and the partition plate is rotatably or detachably arranged in the storage body.

[0023] In some embodiments, the refrigeration assembly includes a compressor, a condenser pipe connected to the compressor, an evaporator pipe connected to the condenser pipe, and a cooling fan arranged on one side of the condenser pipe, and the refrigeration base is provided with at least one pair of ventilation windows for dissipating heat of the

refrigeration assembly; one pair of ventilation windows are arranged on two opposite surfaces of the refrigeration base; or one pair of ventilation windows are arranged on two adjacent surfaces of the refrigeration base; and the soft outer package wraps the refrigeration base in a way that avoids the ventilation windows.

[0024] In some embodiments, the refrigeration base has a mounting surface, the battery mounting portion is arranged on the mounting surface, and the mounting surface is not coplanar with the two ventilation windows; the one pair of ventilation windows are arranged on two opposite surfaces of the refrigeration base, and the mounting surface is perpendicular to a plane where the two ventilation windows are located.

[0025] In some embodiments, the temperature control module further includes a control circuit, the control circuit is arranged inside the refrigeration base, the battery mounting portion is electrically connected to the control circuit, the control circuit is electrically connected to the refrigeration assembly, an outer side of the refrigeration base is provided with a control panel, the control panel is electrically connected to the control circuit, and the soft outer package wraps the refrigeration base in a way that avoids the ventilation windows.

[0026] In some embodiments, the battery mounting portion has an opening for removing the portable power source, and the opening enables the portable power source to be inserted into the battery mounting portion in a vertically downward direction.

[0027] In some embodiments, the temperature-controlled storage device further includes a carrying assembly, the carrying assembly is detachably connected to the soft outer package and/or the storage module for the user to carry the temperature-controlled storage device conveniently; and the carrying assembly includes a human carrying part, and the human carrying part is detachably connected to the soft outer package and/or the storage module.

[0028] In some embodiments, the temperature-controlled storage device further includes a detachable tabletop, and the tabletop can be partially supported and unfolded relative to the storage module; or the temperature-controlled storage device includes a detachable bracket, and the soft outer package is detachably mounted with a metal bracket for fixing the soft outer package on a vehicle; or the temperature-controlled storage device comprises a detachable wheel assembly, wherein the wheel assembly is detachably connected to the soft outer package, and wheels of the wheel assembly are positioned near a bottom of the temperature-controlled storage device for mobility; or the temperature-controlled storage device comprises a camping expansion assembly, and the camping expansion assembly is disposed on one or more defined by four radial peripheral surfaces of the temperature-controlled storage device for being connected to external accessories.

[0029] In some embodiments, the power supply assembly includes one or more portable power sources, and the portable power source is detachably arranged in the battery mounting portion; the power supply assembly further includes a photovoltaic power panel and the portable power source that electrically connected to the battery mounting portion, the photovoltaic power panel is configured for supplying power to the portable power source; or the power supply assembly further includes at least one power supply interface configured for being electrically connected to an

external power source, such that the external power source supplies power to the temperature control module, or the portable power source inside the battery mounting portion; and types of the power supply interface include TYPE-C, AC female plug, delta-shaped interface, and USB interface.

[0030] In some embodiments, the temperature control module includes the control circuit, and the battery mounting portion is electrically connected to the control circuit; when the compressor of the temperature control module is working, the control circuit controls current inputted to the compressor; and when the compressor of the temperature control module is not driven, the control circuit controls current inputted to the portable power source until the portable power source is fully charged.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a schematic diagram of principles of a temperature-controlled storage device according to one embodiment of the present disclosure.

[0032] FIG. 2 is a schematic diagram of principles of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0033] FIG. 3 is an exploded view of a temperature-controlled storage device according to one embodiment of the present disclosure.

[0034] FIG. 4 is a three-dimensional schematic diagram of a temperature-controlled storage device according to one embodiment of the present disclosure.

[0035] FIG. 5 is a three-dimensional schematic diagram of a temperature-controlled storage device that is provided with a power supply panel according to one embodiment of the present disclosure.

[0036] FIG. 6 is an exploded view of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0037] FIG. 7 is a sectional view of a temperature-controlled storage device according to one embodiment of the present disclosure.

[0038] FIG. 8 is a sectional view of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0039] FIG. 9 is a sectional view of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0040] FIG. 10 is an exploded view of a temperature-controlled storage device that is provided with an expansion sleeve according to one embodiment of the present disclosure.

[0041] FIG. 11 is a three-dimensional schematic diagram of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0042] FIG. 12 is a top view of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0043] FIG. 13 is a sectional view of a temperature-controlled storage device that is combined with an electric vehicle according to one embodiment of the present disclosure.

[0044] FIG. 14A is a three-dimensional schematic diagram of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0045] FIG. 14B is a sectional view of a temperature-controlled storage device that is provided with a hot air circulation vent according to one embodiment of the present disclosure.

[0046] FIG. 15 is a three-dimensional schematic diagram of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0047] FIG. 16 is a schematic diagram of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0048] FIG. 17 is a schematic diagram of the temperature-controlled storage device in FIG. 16 from another angle of view.

[0049] FIG. 18 is a partial sectional view of the temperature-controlled storage device in FIG. 16.

[0050] FIG. 19 is a schematic diagram of a metal bracket of a temperature-controlled storage device according to one embodiment of the present disclosure.

[0051] FIG. 20A is a schematic diagram of an external structure of an overall storage body of a temperature-controlled storage device according to a preferred embodiment of the present disclosure.

[0052] FIG. 20B is a schematic cross-sectional view of the temperature-controlled storage device in FIG. 20A.

[0053] FIG. 21 is a schematic diagram of a temperature control module of the temperature-controlled storage device in FIG. 20A.

[0054] FIG. 22 is a schematic cross-sectional view of a temperature-controlled storage device according to a variant embodiment in FIG. 20A.

[0055] FIG. 23A is a schematic diagram of an external structure of an overall storage body of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0056] FIG. 23B is a schematic cross-sectional view of a temperature-controlled storage device according to another embodiment of the present disclosure.

[0057] FIG. 24 is a schematic diagram of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0058] FIG. 25 is a schematic diagram of the temperature-controlled storage device in FIG. 24 from another angle of view.

[0059] FIG. 26 is a schematic diagram of the temperature-controlled storage device in FIG. 25 from which a soft backpack is removed.

[0060] FIG. 27 is a schematic diagram of a variant embodiment of the embodiment in FIG. 26.

[0061] FIG. 28 is a schematic cross-sectional view of a temperature-controlled storage device according to yet another embodiment of the present disclosure.

[0062] FIG. 29 is a schematic diagram of a temperature-controlled storage device that is provided with a temperature control module according to one embodiment of the present disclosure.

[0063] FIG. 30 is a top sectional view of a temperature-controlled storage device according to one embodiment of the present disclosure.

REFERENCE NUMERALS IN THE ACCOMPANYING DRAWINGS

[0064] 100. storage module; 102. storage body; 104. storage lid; 110. first storage body; 120. second storage body; 127. partition plate; 129. partition plate shifting

member; **131.** partition plate conversion positioning member; **135.** air outlet; **137.** hot air circulation vent; **112.** anti-overflow space; **116.** heating element; **216.** expansion sleeve; **108.** sealing limiting member;

[0065] **200.** soft outer package; **202.** soft pack cover; **204.** soft pack body; **220.** ear pouch; **500.** soft back-pack; **501.** metal bracket; **502.** battery pocket; **503.** through hole; **504.** wheel assembly;

[0066] **250.** temperature control module; **450.** refrigeration assembly; **451.** compressor; **452.** condenser pipe; **453.** cooling fan; **454.** evaporator pipe; **404.** control circuit; **402.** refrigeration base; **4021.** ventilation window; **4022.** control panel; **4023.** external power port; **4024.** wiring hole; **4025.** mounting surface; **118.** cooling element;

[0067] **300.** carrying assembly; **310.** human carrying part; **312.** main body; **314.** shock-absorbing member; **316.** camping expansion assembly; **318.** carrying extension assembly;

[0068] **210.** power supply panel; **213.** power supply bracket; **405.** portable power source; **409.** battery mounting portion; **411.** fixing member; and **407.** power supply interface.

DETAILED DESCRIPTIONS OF THE EMBODIMENTS

[0069] The technical solutions of embodiments of the present disclosure will be described below clearly and comprehensively in conjunction with accompanying drawings of the embodiments of the present disclosure. Apparently, the embodiments described are merely some embodiments rather than all embodiments of the present disclosure. On the basis of the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without making creative efforts shall fall within the scope of protection of the present disclosure.

[0070] Any reference to “example(s)” or “embodiment(s)” herein means that particular features, structures, or characteristic described in conjunction with the example(s) or embodiment(s) may be included in at least one embodiment of the present disclosure. Any phrase appearing in various places in the specification does not necessarily relate to the same embodiment, nor is an independent or alternative embodiment mutually exclusive with other embodiments. It is explicitly and implicitly understood by those skilled in the art that the embodiments described herein may be combined with other embodiments.

[0071] In the specification of the present disclosure, for convenience only, it is to be understood that the terms “central”, “upper”, “lower”, “front”, “back”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, etc. indicating azimuthal or positional relations are used to illustrate the positional relationship of the constituent elements with reference to the drawings and only for ease of description of the present disclosure and for simplicity of description, and are not intended to indicate or imply that the referenced device or element must have a particular orientation and be constructed and operative in a particular orientation, and thus may not be construed as a limitation on the present disclosure. The positional relationship of the constituent elements can be properly changed according to the orientation of the constituent elements described. Therefore, it is

not limited to the words and phrases described in the specification, and can be appropriately replaced according to the situation.

[0072] The temperature-controlled storage device of the present disclosure includes a storage module **100**, a soft outer package **200**, a temperature control module **250** and a power supply assembly. The soft outer package **200** wraps around a periphery of the storage module **100**, at least a portion of a surface of the temperature control module **250** is covered by the soft outer package **200**, and the temperature control module **250** is configured to adjust an internal temperature of the storage module **100** and is configured for cooling when being powered by the power supply assembly.

[0073] Arrangement of the soft outer package **200** has the following advantages: first, the soft outer package can keep the storage module **100** warm and enhance the heat preservation or cooling preservation effect of the temperature-controlled storage device; second, flexible material of the soft outer package **200** can modify appearance of the temperature-controlled storage device, making parts protruding from the storage module less obtrusive, such that an overall shape of the temperature-controlled storage device is not limited to a conventional square shell; third, unlike traditional metal or plastic appearance, the soft outer package **200** creates a sense of visual lightness; fourth, the flexibility of the soft outer package **200** can improve the comfort when a user carries the temperature-controlled storage device, especially when the user carries the temperature-controlled storage device on his/her back, the soft outer package **200** can play a good role of buffer and makes the user more comfortable when carrying the device; and fifth, the temperature control module **250** is wrapped and modified with the soft outer package **200**, so as to prevent the temperature control module **250** from being directly exposed to the outside, which plays a role in protecting the temperature control module **250** and increase an overall aesthetic appeal.

[0074] The power supply assembly includes a battery mounting portion **409** electrically connected to the temperature control module **250**, and a portable power source **405** can be detachably mounted in the battery mounting portion **409** to supply power to the temperature control module **250**; and the battery mounting portion **409** is arranged on an outer side of the soft outer package **200**, or the soft outer package **200** partially wraps around the battery mounting portion **409**, or the soft outer package **200** can wrap around the battery mounting portion **409** in an openable and closable manner, so as to facilitate the replacement of portable power source **405** in the battery mounting portion **409** at any time.

[0075] At least a portion of the temperature control module **250** protrudes from a radial perimeter surface of the storage module **100**, and/or at least a portion of the power supply assembly protrudes from the radial perimeter surface of the storage module **100**. In other words, the temperature control module **250** can at least partially protrude from the radial perimeter surface of the storage module **100**, as shown in FIGS. 1 and 2; or at least a portion of the power supply assembly protrudes from the radial perimeter surface of the storage module **100**, as shown in FIGS. 3-12; or both the temperature control module **250** and the power supply assembly are at least partially protrude from the radial perimeter surface of the storage module **100**, as shown in FIGS. 26, 27, and 28. It is worth mentioning that the radial perimeter surface of the storage module **100** mentioned in

the present disclosure refers to a surface excluding top and bottom surfaces, and can also be referred to as a circumferential surface.

[0076] In the prior art, in order to maintain an overall square shape of the temperature-controlled storage device, the temperature control module and the power supply assembly are usually embedded into an accommodation space of the storage module, which has an advantage that a regular shape can be obtained, but it has a disadvantage that a storage space of the storage module will be reduced. In order to increase the storage space of the storage module, a length, width or height of the storage module must be increased as a whole, which will inevitably increase overall dimensions of the device, making it less compact and reducing portability of the device. In the present disclosure, at least one of the temperature control module 250 and the power supply assembly protrudes from the radial perimeter surface of the storage module. Although the shape is no longer a regular rectangle, more space can be left inside the storage module 100 for storage purposes, thereby increasing a storage volume while maintaining the miniaturization of the storage module 100. Considering that the soft outer package 200 introduced in the present disclosure can also modify the irregular shape, the protrusion of the temperature control module 250 and the power supply assembly will not make the overall shape appear uncoordinated.

[0077] In some embodiments, as shown in FIG. 3, the temperature-controlled storage device further includes a carrying assembly 300, the carrying assembly 300 is detachably connected to the storage module 100 or the soft outer package 200 for the user to carry the temperature-controlled storage device conveniently. Further, the carrying assembly 300 can include a human carrying part 310, such as a shoulder strap, and the user can easily carry the temperature-controlled storage device through the human carrying part 310. Preferably, the human carrying part 310 is connected to a back surface of the storage module 100, and neither the temperature control module 250 nor the power supply assembly protrudes from the back surface thereof, so as to ensure comfort when carrying. Further, the radial perimeter surface of the storage module 100 includes two side surfaces adjacent to the back surface and a front surface opposite to the back surface, the temperature control module 250 at least partially protrudes from the side surfaces of the storage module 100, and/or the power supply assembly at least partially protrudes from the side surfaces of the storage module 100.

Storage Module 100

[0078] As shown in FIGS. 1, 2 and 3, the storage module 100 includes a storage body 102 and a storage lid 104 disposed on the storage body 102 in an openable and closable manner, the storage body 102 defines the storage space, and the storage lid 104 is configured to seal or open the storage space. The temperature control module 250 is configured to adjust a temperature in the storage space defined by the storage body 102.

[0079] An opening of the storage body 102 can be arranged on a top surface or a radial perimeter surface thereof, and the storage lid 104 can be arranged at the opening of storage body 102 in an openable and closable manner. In several embodiments illustrated in the accompanying drawings of the present disclosure, the opening of the storage body 102 is arranged on the top surface. The storage

lid 104 can be hinged at the opening of the storage body 102, or can be detachably covered on the opening of the storage body 102, or can be movably arranged at the opening of the storage body 102 in a pull-out manner, and further, the storage lid 104 can be a double-door lid. The present disclosure does not limit the opening and closing form of the storage lid 104; and all types of storage lids 104 fall within the scope of protection of the present disclosure.

[0080] In some embodiments, as shown in FIGS. 8 and 9, an interior of the storage body 102 is divided into different temperature zones. Specifically, the storage body 102 includes a first zone and a second zone that are spaced apart, and the temperature control module 250 can cool the first zone and the second zone to different degrees, such that the first zone and the second zone can maintain different temperatures. Specifically, the storage body 102 is provided with a partition plate 127, the partition plate 127 divides a storage space of the storage body 102 into the first zone and the second zone, an air outlet 135 is formed on an inner wall of the storage body 102, and the air outlet 135 is located in the first zone, such that a temperature in the first zone is lower than that in the second zone. The partition plate 127 can be fixedly arranged in the storage body 102, as shown in FIG. 9, or can be movably arranged in the storage body 102 to adjust spaces of the first zone and the second zone as needed, as shown in FIG. 8, or the partition plate 127 can be detachably arranged in the storage body 102, allowing the user to choose whether to partition the space according to actual needs of the user.

[0081] Further, as shown in FIG. 6, 7, 8, 9, 14, 20B, 22, 23A, 23B, 27, 28 or 29, a bottom of the storage body 102 is non-planar, a portion of the bottom surface of the storage body is recessed to form a concave space to accommodate the temperature control module 250 and/or the power supply module.

[0082] In some embodiments, the bottom of the storage body 102 is stepped, such that an interior space of the storage body 102 has two portion, high portion and low portion, as shown in FIGS. 7, 8 and 9. The partition plate 127 can be vertically arranged at a step of the storage body 102, such that the storage space is divided into a left storage space and a right storage space, with one side being a deeper storage area and the other side being a shallower storage area, as shown in FIG. 9, and different items can be placed in the two storage areas to meet various different needs of the user. The partition plate 127 can also be horizontally placed at the step of the storage body 102, such that the storage space is divided into an upper storage space and a lower storage space, with the lower storage space being small and the upper storage space being large. In some embodiments, the partition plate 127 is rotatably arranged on a partition plate shifting member 129, a partition plate conversion positioning member 131 is arranged inside the storage body 102, and the partition plate conversion positioning member 131 can make the partition plate 127 to stay at different positions; when the partition plate 127 stays at the different positions, the storage space can be selectively divided into a left storage space and a right storage space or an upper storage space and a lower storage space; when the partition plate 127 stays horizontally, the storage space is divided into the upper storage space and the lower storage space, the lower storage space is relatively small, the air outlet 135 is arranged in the lower storage space, such that a temperature of the lower storage space can be reduced

quickly, or the lower storage space can have a lower temperature, thereby achieving temperature zoning management.

[0083] The bottom of the storage body **102** is stepped, that is, one side of the bottom surface of the storage body **102** is recessed inward to form a concave space, as shown in FIG. 6. The temperature control module **250** is arranged in the concave space of the storage body **102**, top and one side surface of the temperature control module **250** are in close contact with the storage body **102**, and the other three side surfaces and a bottom surface of the temperature control module **250** are exposed relative to the storage body **102**, such that a bottom space of the storage body **102** is fully utilized. In some of the embodiments, the temperature control module **250** does not protrude from the radial perimeter surface of the storage body **102**, as shown in FIGS. 7, 8, and 9. In some of the other embodiments, one side or two opposite sides of the temperature control module **250** protrude from the radial perimeter surface of the storage body **102**, as shown in FIGS. 27 and 28. Preferably, the bottom surface of the temperature control module **250** and the bottom surface of the storage body **102** are substantially in a same plane, thus improving the placement stability.

[0084] In other embodiments, only one corner of the bottom of the storage body **102** is recessed inward to form a concave space; when the temperature control module **250** is arranged in the concave space of the storage body **102**, a top and two adjacent side surfaces of the temperature control module are in close contact with the storage body **102**, and a bottom surface and the other two adjacent side surfaces of the temperature control module are exposed relative to the storage body **102**. In some of the embodiments, the temperature control module **250** does not protrude from the radial perimeter surface of the storage body **102**. In other embodiments, one side of the temperature control module **250** protrudes from the radial perimeter surface of the storage body **102**. In other embodiments, two adjacent side surfaces of the temperature control module **250** protrude from two adjacent sides of the radial perimeter surface of the storage body **102**.

[0085] In other variant embodiments, a bottom surface of the storage body **102** is a plane, the temperature control module **250** and the storage body **102** arranged in parallel on one side of the storage body **102**, as shown in FIG. 26, and a bottom surface of the temperature control module **250** and a bottom surface of the storage body **102** are in the same plane.

[0086] In some embodiments, the storage body **102** includes a first storage body **110** and a second storage body **120** that is sleeved and arranged in the first storage body **110**, as shown in FIG. 3, the first storage body **110** and the second storage body **120** are arranged at intervals, insulation material such as polyurethane foam, polystyrene board, phenolic foam, glass wool, rock wool, and other material with heat-insulating properties, can be placed between the first storage body and the second storage body, or vacuum can be maintained between them. Further, the storage body **102** includes a sealing edge **125** connected to upper ends of the first storage body **110** and the second storage body **120**.

[0087] In some embodiments, an expansion sleeve **216** is arranged between the storage body **102** and the storage lid **104**, as shown in FIG. 10, two ends of the expansion sleeve **216** are connected to the storage body **102** and the storage lid **104**, respectively. By introducing the expansion sleeve

216, a storage space of the temperature-controlled storage device can be expanded. The two ends of the expansion sleeve **216** can be directly connected to form an upper and lower communicating cylinder, such that a space defined by the expansion sleeve **216** is communicated with a space defined by the storage body **102**. In some other embodiments, the expansion sleeve **216** is provided with a bottom, forming a compartment with a bottom, meaning that the space defined by the expansion sleeve **216** is separated from the space defined by the storage body **102**.

[0088] A sealing limiting member **108** can be configured to seal one end of the expansion sleeve **216** and the storage body **102**, and to seal the other end of the expansion sleeve **216** and the storage lid **104**. Alternatively, other sealing methods can also be adopted, such as rotary connection with a sealing strip or concave and convex limiting with a sealing strip.

[0089] In some embodiments, the storage lid **104**, the storage body **102**, and the expansion sleeve **216** are all provided with fabric outer covers. A plurality of the fabric outer covers are connected by zippers to form a soft backpack **500**.

[0090] In some embodiments, the storage lid **104** can have different height dimensions, and by replacing different storage lids **104**, the temperature-controlled storage device can have different storage spaces.

Soft Outer Package 200

[0091] As shown in FIG. 1, the soft outer package **200** includes a soft pack body **204**, the soft pack body **204** wraps the storage body **102**, and the soft pack body **204** covers at least one side of the temperature control module **250**. The soft pack body **204** and the storage body **102** can be assembled as a single unit or integrally formed, or can be arranged separately. Preferably, the soft pack body **204** is assembled as a single unit or integrally formed with the storage body **102**, which is conducive to improving the heat preservation effect and the compactness of the temperature-controlled storage device.

[0092] In some embodiments, as shown in FIG. 2 or 3, the soft outer package **200** further includes a soft pack cover **202**, and the soft pack cover **202** wraps the storage lid **104**. The soft pack cover **202** and the storage lid **104** can be either assembled as a single unit or integrally molded, or they can be arranged separately. Preferably, the soft pack cover **202** is assembled as a single unit or integrally molded with the storage lid **104**, which helps improve insulation and enhances the compactness of the temperature-controlled storage device.

[0093] In some embodiments, the soft pack cover **202** and the soft pack body **204** are arranged separately, such that the soft pack cover **202** can be separated from the storage body **102** and the soft pack body **204** along with the storage lid **104**. In other embodiments, the soft pack cover **202** is connected to the soft pack body **204**.

[0094] The soft outer package **200** can be made from fabric, leather, and the like, which is configured to modify the appearance of the temperature-controlled storage device on the one hand, and can improve the tactile feel and user comfort during carrying on the other hand.

[0095] In order to improve the airtightness of the storage module **100**, the storage module **100** includes a sealing limiting member **108** for sealing the connection between the storage lid **104** and the storage body **102**, as shown in FIG.

3 or 14A. In some embodiments, the sealing limiting member 108 is a metal latch, which provides better sealing performance for the storage module 100 and also facilitates easy opening of the storage lid 104 for retrieval of items. Two ends of the sealing limiting member 108 are respectively connected to the soft pack body 204 and the soft pack cover 202, as shown in FIG. 14A; or the two ends of the sealing limiting member 108 are connected to the storage body 102 and the storage lid 104, as shown in FIG. 3; or the two ends of the sealing limiting member 108 are connected to the storage body 102 and the soft pack cover 202; or the two ends of the sealing limiting member 108 are connected to the soft pack body 204 and the storage lid 104. A number of the sealing limiting member 108 can be one, two, or more. In some variant embodiments, the sealing limiting member 108 can also be a layer of edging, the storage module 100 can be opened and closed by setting a zipper on the edging, or sealing effect can be achieved by arranging concave and convex portions on the soft outer package 200 and the storage lid 104.

[0096] In some embodiments, the storage body 102 and the storage lid 104 define a sealing connection position, while the soft pack body 204 and the soft pack cover 202 define an accommodating connection position, the sealing connection position and the accommodating connection position are arranged in a staggered manner in a height direction, as shown in FIG. 3, that is, the connection positions of inner and outer layers are staggered, which is conducive to improving the heat preservation effect. In other embodiments, the sealing connection position and the accommodating connection position are at a same level, and this design mainly gives consideration to processing cost and convenience of use.

[0097] In some embodiments, the storage body 102 and the storage lid 104 define a sealing connection position, a top of the soft pack body 204 extends upward beyond the sealing connection position, or a bottom of the soft pack cover 202 extends downward beyond the sealing connection position (as shown in FIG. 2), such that the heat preservation effect is improved.

[0098] As shown in FIGS. 1, 2, and 3, the soft outer package 200 further includes at least one ear pouch 220, the ear pouch 220 is arranged on a radial perimeter surface of the storage module 100. The ear pouch 220 can be configured to cover the battery mounting portion 409 of the power supply assembly, as shown in FIG. 3, or to cover other components protruding from the radial perimeter surface of the storage module 100, for example, to cover the temperature control module 250, as shown in FIGS. 1 and 2, or to serve as a side pocket for storage, as shown in FIG. 14B. The ear pouch 220 can be configured to modify the components protruding from the radial perimeter surface of the storage module 100. In some embodiments, the ear pouch 220 covers the battery mounting portion 409, and the ear pouch 220 is provided with a corresponding opening to facilitate the removal of a battery. The arrangement of the ear pouch 220 significantly alters the appearance of the temperature-controlled storage device, making it different from the metal appearance of traditional temperature control devices, thereby creating a sense of visual lightness and actually providing a higher volume-to-weight ratio.

[0099] It is worth mentioning that the soft outer package 200 refers to soft material capable of covering and accommodating the storage module 100, the soft outer package has

a moldable form and can modify the appearance of the temperature-controlled storage device, such that the temperature-controlled storage device is not necessarily a square casing.

[0100] As shown in FIG. 3, an anti-overflow space 112 is arranged between the storage body 102 and the soft outer package 200, and the temperature control module 250 is arranged in the anti-overflow space 112.

[0101] In some embodiments, as shown in FIG. 7, in order to enable the temperature-controlled storage device to heat food, the anti-overflow space 112 further includes at least one heating element 116, the heating element 116 is electrically connected to a control circuit 404, the arrangement has the advantages that food stored in the storage module 100 can be heated in cold weather, so as to cope with various outdoor conditions. The heating element 116 can be a heating wire.

[0102] It is worth mentioning that when wrapping the temperature control module 250, the soft outer package 200 does not wrap the temperature control module 250 completely in a sealed manner. For example, when the temperature control module 250 is provided with a ventilation window 4021 for heat dissipation, as shown in FIG. 20A, the soft outer package 200 wraps the temperature control module 250 in a way that avoids the ventilation windows, that is, ensuring that the soft outer package 200 does not block the ventilation window; or when the temperature control module 250 is provided with a control panel 4022 for user operation, as shown in FIG. 20A, the soft outer package 200 wraps the temperature control module 250 in a way that avoids the control panel 4022, that is, ensuring that the soft outer package does not block the control panel 4022.

Temperature Control Module 250

[0103] As shown in FIG. 6, the temperature control module 250 includes a refrigeration base 402 and a refrigeration assembly 450 arranged inside the refrigeration base 402, and the refrigeration base 402 is connected to the storage module 100, such that the refrigeration assembly 450 is close to the storage module 100 to cool the storage module down. The temperature control module 250 further includes a control circuit 404, the control circuit 404 is installed on the refrigeration base 402, the control circuit 404 is electrically connected to the power supply assembly and the refrigeration assembly 450, so as to control the power supplied by the power supply assembly to the refrigeration assembly 450.

[0104] The refrigeration base 402 forms an outer shell of the temperature control module 250 and is connected to the storage body 102. The refrigeration assembly 450 and the control circuit 404 are both arranged inside the refrigeration base 402, the power supply assembly is electrically connected to the control circuit 404, and the control circuit 404 is also electrically connected to the refrigeration assembly 450, such that the power supply assembly supplies power to the refrigeration assembly 450 through the control circuit 404.

[0105] In some embodiments, the refrigeration base 402 is detachably connected to the storage body 102, in other words, the temperature control module 250 is detachably connected to the storage body 102, such that the user can remove the entire temperature control module 250 from the temperature-controlled storage device as needed, reducing a weight for easier portability, while retaining a basic insulation function of the storage module 100.

[0106] In some embodiments, as shown in FIG. 23A, the refrigeration base 402 is provided with an outward-facing mounting surface 4025, and the battery mounting portion 409 of the power supply assembly is arranged on the mounting surface 4025, such that a portable power source 405 can supply power to the refrigeration assembly 450 conveniently. When the refrigeration base 402 is removed from the storage body 102, the battery mounting portion 409 is also removed, further reducing a weight of the remaining parts. The mounting surface 4025 should avoid being arranged at a back of the temperature-controlled storage device, and in order to enable the user to easily take the portable power source 405 when carrying the temperature-controlled storage device, the mounting surface 4025 is located on a side adjacent to the back.

[0107] In other embodiments, the battery mounting portion 409 is not installed on the refrigeration base 402 but on the storage module 100 instead, so as to be kept away from the refrigeration assembly 450 with a great amount of heat generation to avoid affecting the normal use of the battery. For example, the battery mounting portion 409 can be installed on the storage body 102 or on the storage lid 104.

[0108] In some embodiments, as shown in FIG. 21, the refrigeration assembly 450 includes a compressor 451, a condenser pipe 452 connected to the compressor 451, an evaporator pipe 454 connected to the condenser pipe 452, and a cooling fan 453 arranged on one side of the condenser pipe 452, and the condenser pipe 452 is partially exposed to an exterior of the soft outer package 200. As shown in FIGS. 20B and 22, the evaporator pipe 454 can be extended between the first storage body 110 and the second storage body 120 and configured to reduce a temperature of the storage body 102, the evaporator pipe 454 can be tightly wound around an exterior of the second storage body 120, the compressor 451 works with the condenser pipe 452 and the evaporator pipe 454 in a coordinated manner to form a complete refrigeration cycle system, which can efficiently and precisely cools an interior of the storage body 102. The cooling fan 453 is specifically provided to dissipate heat from the condenser pipe 452 to ensure that the cooling process can proceed smoothly. It is worth mentioning that the evaporator pipe 454 is also used as a cooling element 118 in some embodiments of the present disclosure.

[0109] In some embodiments, the control circuit 404 includes at least two modes: including a first cooling mode and a second cooling mode, respectively, the first cooling mode and the second cooling mode can be switched, and the first cooling mode and the second cooling mode can be selected and triggered by a key.

[0110] Further, as shown in FIG. 21, the refrigeration base 402 is provided with at least one pair of ventilation windows 4021, where at least one of the ventilation windows 4021 is configured for air inlet, and the other ventilation window 4021 is configured for air outlet; and the cooling fan 453 and the condenser pipe 452 are arranged near one of the ventilation windows 4021. At least one pair of ventilation windows 4021 facilitates the flow of external air and defines a heat dissipation channel, and all components of the refrigeration assembly 450 are arranged along the heat dissipation channel to improve the heat dissipation effect. In some embodiments, one pair of ventilation windows 4021 are arranged on two opposite surfaces of the refrigeration base

402. In other embodiments, one pair of ventilation windows 4021 are arranged on two adjacent surfaces of the refrigeration base 402.

[0111] In some embodiments, the refrigeration base 402 is provided with two ventilation windows 4021, the mounting surface 4025 of the refrigeration assembly 450 is not coplanar with the ventilation windows 4021, as shown in FIG. 23A. For example, the two ventilation windows 4021 are arranged on two opposite surfaces of the refrigeration assembly 450, one of the ventilation windows is located on a front side of the temperature-controlled storage device, and the other is located on a back side of the temperature-controlled storage device; the mounting surface 4025 is perpendicular to a plane where the two ventilation windows 4021 are located, that is, the mounting surface 4025 is located on a side surface of temperature-controlled storage device, where the ventilation window 4021 on the front side is configured for air outlet, the ventilation window 4021 on the back side is configured for air inlet to avoid blocking the ventilation windows 4021 for discharging heat when the user carries the temperature-controlled storage device. Further, the cooling fan 453 is arranged near the ventilation window 4021 for air outlet.

[0112] In some embodiments, one side of the refrigeration base 402 is provided with a control panel 4022, as shown in FIG. 23A, and the control panel 4022 is electrically connected to the control circuit 404. The control panel 4022 not only provides an intuitive operating interface for the user but also allows the user to conveniently perform various operations on the refrigeration assembly 450. Through the control panel 4022, the user can easily adjust cooling parameters, monitor the operating status, and even perform fault diagnostics and repair, greatly improving the usability and maintainability of the device.

[0113] The control panel 4022 can be arranged on a surface same as that of the ventilation windows 4021 or on a side without the ventilation windows 4021. Further, the control panel 4022 can be also arranged on the mounting surface 4025 or on a side opposite to the mounting surface 4025. In some preferred embodiments, the control panel 4022 is coplanar with the ventilation window 4021 for air outlet, that is, the control panel 4022 faces a front side.

[0114] In one embodiment of the ventilation window 4021, a first ventilation window 4021 is arranged on a side of the refrigeration base 402 facing a front side or a back side, a second ventilation window 4021 is arranged on a side of the refrigeration base 402 facing a side, the two ventilation windows 4021 are respectively set as an air inlet end and an air outlet end, and the condenser pipe 452 is near the air outlet end.

[0115] In another embodiment of the ventilation window 4021, as shown in FIG. 30, the ventilation window 4021 protrudes from the refrigeration base 402, such that an installation area of the ventilation window 4021 can be expanded, the cooling fan 453 can be installed inside the ventilation window 4021, making an interior of the refrigeration base 402 more spacious and facilitating the airflow, thereby improving the heat dissipation effect. Of course, the protruding ventilation window 4021 also allows for the addition of the cooling fan 453 inside the protruding ventilation window 4021 to increase the exhaust volume, such that the heat dissipation effect of the temperature control module 250 can also be improved.

[0116] In one embodiment of the ventilation window 4021 (not shown in the figure), three or more ventilation windows 4021 can be arranged, and each of the ventilation windows 4021 is not arranged on a same surface; where at least one of the ventilation windows 4021 is set as the air inlet end, at least one of the ventilation windows 4021 is set as the air outlet end, and the remaining ventilation windows 4021 are set as either air inlet end or air outlet end depending on actual needs, such that the ventilation of air volume can be increased, and the heat dissipation and ventilation effects can be improved. Of course, in order to increase the airflow of air inlet or air outlet on one side, a plurality of coplanar ventilation windows 4021 can be installed on a single side, which has the same effect as that of increasing the area of the ventilation windows 4021.

[0117] In a variant embodiment (not shown in the figure), at least one side of the refrigeration base 402 is set as a stepped surface, such that a number of sides of the refrigeration base 402 can be increased, and more ventilation windows 4021 can be arranged, thereby further improving the heat dissipation and ventilation effects.

[0118] In another variant embodiment (not shown in the figure), a plurality of adjacent ventilation windows 4021 are all set as air inlet ends, thereby forming a large air inlet window, such that an air volume entering an interior of the refrigeration base 402 is increased, and the heat dissipation and cooling effects are improved. Of course, in other embodiments, a plurality of adjacent ventilation windows 4021 can all be set as air outlet ends, such that an air exhaust volume of the refrigeration base 402 is increased, air convection is accelerated, and the heat dissipation and cooling effects are improved. Of course, in yet other embodiments, a plurality of ventilation windows 4021 can be divided into two groups, one group of adjacent ventilation windows 4021 is set as an air inlet end, and the other group of adjacent ventilation windows 4021 is set as an air outlet end, in such a way, the air inlet volume and the air exhaust volume of the refrigeration base 402 are both increased, and the heat dissipation and cooling effects become better.

[0119] In some embodiments, the control panel 4022 protrudes from a radial perimeter surface of the storage body 102, an opening of the battery mounting portion 409 is oriented parallel to a heat dissipation channel defined by at least two ventilation windows 4021, that is, the mounting surface 4025 and the heat dissipation channel defined by the two ventilation windows 4021 are arranged in a same direction.

[0120] In a variant embodiment shown in FIGS. 26 and 27, a center of gravity (G) of the temperature control module 250 is far away from the storage body 102. The center of gravity (G) of the temperature control module 250 is located in a range of 1-1.5 times a radius (R) of a central axis of the storage body 102. More preferably, the center of gravity (G) of the temperature control module 250 is located in a range of 1.1-1.3 times the radius (R) of a central axis of the storage body 102.

[0121] The temperature control module 250 and the battery mounting portion 409 are arranged on a same side, and arranged in an axial direction, where a portion of the temperature control module 250 protrudes from the radial perimeter surface of the storage body 102, while the battery mounting portion 409 is installed on the radial perimeter surface of the storage body 102.

Carrying Assembly 300

[0122] In order to facilitate the carrying of the temperature-controlled storage device, the temperature-controlled storage device provided by the present disclosure further includes a carrying assembly 300, as shown in FIG. 3. In some embodiments, the carrying assembly 300 is detachably mounted on the soft outer package 200. The carrying assembly 300 allows the user to conveniently carry or wear the temperature-controlled storage device, and the carrying assembly 300 can also be removed when it is unnecessary to be carried, to facilitate separate use of the storage module 100. The introduction of the carrying assembly 300 improves the portability of the temperature-controlled storage device in the present disclosure, making the temperature-controlled storage device more configured for outdoor use.

[0123] Further, as shown in FIG. 3, a carrying assembly 300 includes a human carrying part 310, the human carrying part 310 includes a main body 312 and a shock-absorbing member 314, the main body 312 has an outer surface and an inner surface, the outer surface of the main body 312 is connected to the storage module 100, and the shock-absorbing member 314 is arranged on the inner surface of the main body 312. The user can carry the temperature-controlled storage device on his/her back via the human carrying part 310, and the introduction of the shock-absorbing member 314 is conducive to improving comfort during carrying.

[0124] In some embodiments, as shown in FIG. 3, the human carrying part 310 is provided with a camping expansion assembly 316, which can be a structure such as a hook. The camping expansion assembly 316 is arranged on the main body 312. By adding the hook, an outdoor item such as a blanket, a tent, and a walking pole, can be arranged, thereby greatly expanding the functionality of the temperature-controlled storage device. Camping indicates movement following movement of the temperature-controlled storage device.

[0125] Alternatively, referring to FIG. 15 of the accompanying drawings, the camping expansion assembly 316 is removably mounted to the soft outer package 200, the camping expansion assembly 316 is disposed on one or more defined by four radial peripheral surfaces of the temperature-controlled storage device, and the camping expansion assembly 316 is used for being connected to external accessories such as power tools, removable table boards, and camping tools. Preferably, the camping expansion assembly 316 is disposed at one or more of the right angles defined by the four radial peripheral surfaces of the temperature-controlled storage device. More preferably, the camping expansion assembly 316 is disposed at four right angles defined by the four radial peripheral surfaces of the temperature-controlled storage device. The camping expansion assembly 316 can be connected or used in combination.

[0126] The temperature-controlled storage device further comprises a carrying assembly 300, which is detachably connected to the storage module 100 and/or the soft outer package 200, for facilitating the user's portability of the temperature-controlled storage device. In some embodiments, unlike the embodiment shown in FIG. 3, the carrying assembly 300 may include a detachable wheel assembly, which is detachably connected to the storage module 100 and/or the soft outer package 200, allowing the wheel assembly to keep the temperature-controlled storage device walking in a generally vertical direction. For example, the

wheel assembly includes at least one pair of wheels and a connector attached to the temperature-controlled storage device. The wheels are positioned near the bottom of the temperature-controlled storage device. The wheels maintain the temperature-controlled storage device in a generally vertical direction for easy movement, allowing the user to carry the temperature-controlled storage device with ease.

[0127] Alternatively, for example, the wheel assembly further includes at least one pair of wheels, a connector attached to the temperature-controlled storage device, and an extendable pull rod, with the wheels positioned near the bottom of the temperature-controlled storage device. The user can pull the temperature-controlled storage device across the ground using the pull rod, making it easier to carry the device while walking. More preferably, the connector connected to the temperature-controlled storage device is a support 501, with reference to FIG. 15. That is, the wheel assembly 504 includes the at least one pair of wheels, the support 501 connected to the temperature-controlled storage device, and the extendable pull rod.

[0128] In some embodiments, as shown in FIG. 13, the carrying assembly 300 includes at least one carrying extension assembly 318, and the carrying extension assembly 318 is arranged on an outer surface of the soft outer package 200 for buckling, hooking, or binding.

[0129] In another embodiment, as shown in FIGS. 16-19, an outside of the storage body 102 is provided and sleeved with a soft backpack 500. In this embodiment, the soft backpack 500 is equivalent to the soft outer package 200 in other embodiments. The soft backpack 500 is provided with a metal bracket 501 for fixing the soft backpack 500 onto vehicle such as a bicycle or car, such that the temperature-controlled storage device can be installed on the vehicle for carrying. By arranging the soft backpack 500 on the storage body 102, a portable solution is provided for the user, the soft backpack 500 can perfectly fit a shape of the storage body 102 and provide good protection for the storage body. The metal bracket 501 can effectively fix the soft backpack 500 on various vehicle, such as bicycles, or cars, such that the user can easily install the storage device on the vehicle, facilitating easy carrying and movement, in such a way, the storage device can be easily carried by the user to meet the storage needs at any time.

[0130] A through hole 503 corresponding to the ventilation window 4021 is formed on the soft backpack 500, as shown in FIG. 16. The soft backpack 500 is provided with a plurality of side pockets, one of the side pockets is a battery pocket 502, the battery pocket 502 is arranged on a side face of a heat dissipation channel defined by two ventilation windows, and the through hole 503 corresponding to the ventilation windows 4021 is provided to ensure that the refrigeration assembly 450 can dissipate heat smoothly.

[0131] In this embodiment, when the outside of the storage body 102 is provided and sleeved with the soft backpack 500, the battery mounting portion 409 and the portable power source 405 can be placed inside the battery pocket 502 for storage and installation, wires can be kept among the storage module 100, the refrigeration base 402 and the storage body 102, and the wires can be covered by the soft backpack 500, such that the portable power source 405 and the wires are protected. The battery pocket 502 is provided with a zipper opening, which facilitates opening and closing

the battery pocket 502 and allows for easy replacement of the portable power source 405 inside the battery pocket 502.

Power Supply Assembly

[0132] As shown in FIG. 3, the power supply assembly includes a battery mounting portion 409 and a portable power source 405 detachably arranged inside the battery mounting portion 409, the battery mounting portion 409 has an opening through which the portable power source 405 can be easily taken, facilitating replacement of the portable power source 405 at any time. The battery mounting portion 409 is electrically connected to the control circuit 404 of the temperature control module 250, such that the temperature control module 250 can obtain power of the portable power source 405 to operate.

[0133] The battery mounting portion 409 protrudes from the storage module 100, which is not only convenient for the user to replace the battery or the portable power source 405, but also conducive to increasing a storage space of the storage module 100.

[0134] The portable power source 405 can be fixed onto the battery mounting portion 409 by assembly or through gravity, or the battery mounting portion 409 can be provided with a fixing member 411, as shown in FIG. 6, the fixing member 411 can reinforce a connection between the portable power source 405 and the battery mounting portion 409, preventing the portable power source 405 from accidentally falling out of the battery mounting portion 409. The fixing member 411 can also be a strap, nylon buckle, and the like, the fixing member 411 can be arranged as two parts that match each other on the portable power source 405 and the battery mounting portion 409.

[0135] In some embodiments, an opening direction of the battery mounting portion 409 faces upward, so as to improve the stability of the portable power source during installation and enable the user to remove the portable power source easily.

[0136] The portable power source 405 can be a lithium-ion battery pack, but is not limited thereto, the portable power source 405 can also be a polymer lithium-ion battery, an alkaline battery, a fuel cell, a hand-operated power generator, or any other power source that meets the power needs.

[0137] The portable power source 405 can be repeatedly charged and discharged, but is not limited thereto; and in other embodiments, the portable power source 405 can be designed for single-use only.

[0138] In some embodiments, the soft outer package 200 has a radial perimeter surface, a top surface, and a bottom surface, the battery mounting portion 409 can be arranged on the radial perimeter surface of the soft outer package 200, on the bottom surface of the soft outer package 200, or on a top surface of the soft outer package 200, that is, on a top surface of the soft pack cover 202. Alternatively, the battery mounting portion 409 may be arranged inside the soft pack cover 202, the battery mounting portion 409 has a battery compartment and a plug-in opening, the plug-in opening is communicated with the battery compartment and an external space to accommodate the portable power source 405.

[0139] The portable power source 405 can be plugged into the battery mounting portion 409 in a direction close to the storage module 100. The portable power source 405 and the battery mounting portion 409 can be arranged on either front or side of the storage module 100, and the portable power source 405 is inserted into the battery mounting portion 409

in a vertical downward or horizontal direction; and the portable power source 405 and the battery mounting portion 409 can also be arranged on a top surface of the storage module 100, and the portable power source 405 can be inserted into the battery mounting portion 409 in a direction parallel to the top surface.

[0140] In some embodiments, the battery mounting portion 409 is connected to the refrigeration base 402 of the temperature control module 250. The battery mounting portion 409 can be integrally formed with the refrigeration base 402, or can be detachably installed on the refrigeration base 402. The battery mounting portion 409 can protrude from the radial perimeter surface of the refrigeration base 402, or does not protrude from the radial perimeter surface of the refrigeration base 402. In this embodiment, in order to improve the heat dissipation effect of the battery mounting portion 409, a ventilation hole can be formed on an outer wall of the battery mounting portion 409 to facilitate the dissipation of heat from the portable power source 405; additionally, a cooling fan can be arranged at a location of the ventilation hole to further improve the heat dissipation effect of the portable power source 405. Alternatively, the battery mounting portion 409 can be arranged near the ventilation window 4021 for air inlet.

[0141] Specifically, the battery mounting portion 409 is mounting on the mounting surface 4025 of the refrigeration base 402. The battery mounting portion 409 and the control circuit 404 are arranged on both sides of the refrigeration assembly 450. The battery mounting portion 409 is arranged near the control panel 4022, and both the battery mounting portion 409 and the control panel 4022 are kept near a corner of the refrigeration base 402. That is to say, the battery mounting portion 409 and the control panel 4022 are arranged and kept on two adjacent surfaces of the refrigeration base 402. The battery mounting portion 409 and the control panel 4022 are perpendicular to each other. The battery mounting portion 409 is mounted on the mounting surface 4025 and is close to the ventilation window 4021 at the air inlet end, such that wires of the battery mounting portion 409 can directly pass through a wiring hole 4024, and are connected to the control circuit 404. The mounting method can minimize wiring distance and length of the wires, making the storage device more compact. By placing the battery mounting portion 409 near the ventilation window 4021 at the air inlet end, the portable power source 405 can be kept as far as possible from the ventilation window 4021 at the air outlet end, thereby minimizing the impact of heat on the portable power source 405. An external power port 4023 and the battery mounting portion 409 are arranged in a same plane.

[0142] In other embodiments, the battery mounting portion 409 is connected to the storage module 100, thereby being kept away from the refrigeration assembly 450, this layout is intended to effectively prevent the heat generated by the refrigeration assembly 450 during operation from negatively affecting the performance of the portable power source 405, through this structure, the portable power source 405 can maintain a more stable and optimal working environment, thereby ensuring the long-term reliable operation of the portable power source. For example, the battery mounting portion 409 in which the portable power source 405 is mounted can be arranged on a side of the storage module 100 away from the temperature control module 250, as shown in FIGS. 18, 20A, and 20B, alternatively, the

battery mounting portion 409 in which the portable power source 405 is mounted can be arranged on the side of the storage module 100 near the temperature control module 250, but located above the temperature control module 250, as shown in FIG. 26. The temperature control module 250 is connected to the portable power source 405 in the battery mounting portion 409, such that the portable power source placed in the battery mounting portion 409 can be less affected by the heat from the temperature control module 250. Due to the modification of the soft outer package 200, wiring of the battery mounting portion 409 can be exposed to an exterior of the storage body 102 but will not be exposed to the user. Another mounting method for the battery mounting portion 409 in which the portable power source 405 is mounted is to mount the battery mounting portion on a top of the storage lid 104, as shown in FIG. 22, such that the portable power source 405 can be kept away from the refrigeration assembly 450 at a maximum distance, thereby further reducing the impact of heat on the performance of the portable power source 405.

[0143] In other embodiments, the battery mounting portion 409 is at least partially covered by the soft outer package 200. For example, the soft outer package 200 can accommodate the battery mounting portion 409 and the portable power source 405 in an openable and closable manner. In a first state, the portable power source 405 is covered by the soft outer package 200. In a second state, the soft outer package 200 can be opened, and the portable power source 405 is then exposed.

[0144] In some embodiments, when being arranged in the battery mounting portion 409, the portable power source 405 is at least partially exposed to an external environment of the soft outer package 200, such that the user can easily take the portable power source.

[0145] In the embodiment shown in FIG. 4, each of the portable power source 405 and the battery mounting portion 409 is provided with one and arranged on one side of the storage module 100, such that an overall appearance is similar to a side pocket of a backpack.

[0146] In some embodiments, as shown in FIG. 28, at least a portion of the temperature control module 250 protrudes from the radial perimeter surface of the storage module 100, and at least a portion of the battery mounting portion 409 also protrudes from the radial perimeter surface of the storage module 100, and portions of the temperature control module 250 and the battery mounting portion 409 that protrude from the radial perimeter surface of the storage module 100 are symmetrically located on both sides of the storage module 100, such that an overall appearance of the temperature-controlled storage device more symmetrical, and a center of gravity of the storage device is balanced, thereby improving the comfort when carrying the storage device.

[0147] As shown in FIG. 5, another embodiment of the power supply assembly is provided, the power supply assembly includes a power supply bracket 213 and a power supply panel 210 mounted on the power supply bracket 213, the power supply panel 210 can be a photovoltaic power panel, and the power supply panel 210 supplies power to the temperature-controlled storage device by absorbing solar energy; and the power supply panel 210 can also provide sunlight shading for the user. The power supply bracket 213 is connected to the storage module 100, or the soft outer

package 200, or the carrying assembly 300. The power supply panel 210 is configured for supplying power to the portable power source 405.

[0148] In some embodiments, as shown in FIG. 6, the power supply assembly includes at least one power supply interface 407, the power supply interface 407 can be used for other power sources to directly supply power to the temperature-controlled storage device. Types of the power supply interface 407 include TYPE-C, AC female plug, delta-shaped interface, and USB interface.

Variant Embodiment 1

[0149] In some variant embodiments, the temperature-controlled storage device further includes a carrying member, and the carrying member can be a rope. The carrying member is detachably mounted on the temperature-controlled storage device in the present disclosure, and the temperature-controlled storage device can be kept from a ground through the carrying member. The soft outer package 200 is provided with a carrying connection part, and the carrying connection part is connected to the carrying member.

Variant Embodiment 2

[0150] In some variant embodiments, a temperature-controlled storage device capable of being assembling is provided, the temperature-controlled storage device further includes an electric vehicle, as shown in FIG. 13. The temperature-controlled storage device provided in the present disclosure is arranged on the electric vehicle, the electric vehicle includes a detachable portable power source, and the portable power source can supply power to both the electric vehicle and the temperature-controlled storage device.

[0151] The electric vehicle and the temperature-controlled storage device provided in the present disclosure are simultaneously powered by the portable power source 405, at least one of the electric vehicle and the temperature-controlled storage device is powered by means of a power cable connection, or at least one of the electric vehicle and the temperature-controlled storage device is directly connected to the power supply by means of electrical connector docking; or at least one of the electric vehicle and the temperature-controlled storage device electrical connectors is powered by means of wireless charging.

[0152] The portable power source 405 can be a battery pack with a plurality of batteries, a battery pack group with a plurality of battery packs, or an energy storage power supply with an inverter circuit.

[0153] Hooks and buckles matching the temperature-controlled storage device can be reserved on the electric vehicle, such that the temperature-controlled storage device in the present disclosure can be detachably fixed on the electric vehicle through the carrying extension assembly 318.

Variant Embodiment 3

[0154] In some variant embodiments, as shown in FIG. 14B, the soft outer package 200 is provided with a hot air circulation vent 137, and the hot air circulation vent 137 is arranged inside the ear pouch 220, and items can be placed in the ear pouch 220. When the temperature-controlled storage device is started, hot air is outputted into the ear pouch 220 to heat the items therein. Hot air recovery or hot air from the fan is used for heat preservation, which can

reduce the energy consumption of the temperature-controlled storage device, save energy and reduce load of the storage device, making it environmentally friendly.

[0155] In some optional embodiments, the temperature-controlled storage device includes a detachable tabletop, and the tabletop can be partially supported and unfolded relative to the storage module 100.

[0156] In some optional embodiments, the temperature-controlled storage device can also be powered by a car's power source, such that a car owner can enjoy cold or warm beverages anytime and anywhere, greatly improving the quality of life in the car.

[0157] In some optional embodiments, the temperature-controlled storage device can meet the requirement of "power source" in spite of being taken off from the car through the on-board power source technology. After reaching a set temperature, the temperature-controlled storage device will stop operation temporarily, such that power consumption for continuous operation for 24 hour is often less than one kilowatt-hour.

[0158] The temperature-controlled storage device provided in the present disclosure has multiple functions, including but not limited to refrigeration, heating, heat preservation, and air cooling functions. The temperature-controlled storage device in the present disclosure mainly supplies power through energy storage technology. By using energy storage for power source, the temperature-controlled storage device has many advantages, including but not limited to the fact that the device can be used in places without power source, such as outdoor activities, picnics, camping, and the like. In the event of a power outage or other emergencies, the temperature-controlled storage device with energy storage can continue to operate, ensuring the freshness of food and beverages. In addition, the temperature-controlled storage device can store electrical energy when power demand is low and release electrical energy when power demand is high, thus achieving efficient energy utilization and other benefits.

[0159] The design of the temperature-controlled storage device provided in the present disclosure can be applied in various fields, including but not limited to outdoor activities such as camping, picnics, and barbecuing. The temperature-controlled storage device offers refrigeration and insulation functions to keep food and beverages at an appropriate temperature. During a long-distance travel, the temperature-controlled storage device can provide travelers with cold drinks and fresh food. In sports events, the temperature-controlled storage device can provide athletes with cold beverages to help them recover their strength. The temperature-controlled storage device can also be used to transport medical products, such as vaccines and blood samples, which need to be stored at specific temperatures. In addition, the temperature-controlled storage device can be used by delivery personnel to transport refrigerated food or beverages. In some experiments that require specific temperatures, the temperature-controlled storage device can provide a stable temperature environment and be used in various fields.

[0160] What is shown and described above is about basic principles, essential features and advantages of the present disclosure. Those skill in the art shall understand that the present disclosure is not limited by the above embodiment, what is described in the above embodiment and the specification is only the principle of the present disclosure, on the

premise of not deviating from the spirits and the scope of the present disclosure, the present disclosure may have various changes and improvements, and these changes and improvements shall fall within the scope requiring protection of the present disclosure. The scope requiring protection of the present disclosure is defined by the appended claims and equivalents thereof.

What is claimed is:

1. A temperature-controlled storage device, comprising a storage module and a temperature control module, wherein when being powered, the temperature control module is configured for cooling to adjust a temperature inside the storage module; wherein the temperature-controlled storage device further comprises a soft outer package, the soft outer package wraps a periphery of the storage module, and at least a portion of a surface of the temperature control module is wrapped by the soft outer package.

2. The temperature-controlled storage device according to claim 1, further comprising a power supply assembly, wherein the power supply assembly comprises a battery mounting portion electrically connected to the temperature control module; a portable power source is configured for being detachably mounted in the battery mounting portion to supply power to the temperature control module; and

the battery mounting portion is arranged on an outer side of the soft outer package, or the soft outer package partially wraps the battery mounting portion, or the soft outer package wraps the battery mounting portion in an openable and closable manner.

3. The temperature-controlled storage device according to claim 2, wherein at least one of:

at least a portion of the temperature control module protrudes from a radial perimeter surface of the storage module, or

at least a portion of the battery mounting portion protrudes from the radial perimeter surface of the storage module.

4. The temperature-controlled storage device according to claim 3, wherein at least the portion of the temperature control module protrudes from the radial perimeter surface of the storage module, at least the portion of the battery mounting portion protrudes from the radial perimeter surface of the storage module, and the portions of the temperature control module and the portion of the battery mounting portion that protrude from the radial perimeter surface of the storage module are symmetrically located on both sides of a storage body of the storage module.

5. The temperature-controlled storage device according to claim 1, wherein the storage module comprises a storage body, and a storage lid which is disposed on the storage body in an openable and closable manner; and the temperature control module comprises a refrigeration base and a refrigeration assembly arranged inside the refrigeration base, the refrigeration base is connected to the storage body, and a battery mounting portion of a power supply assembly of the temperature-controlled storage device is electrically connected to the refrigeration assembly.

6. The temperature-controlled storage device according to claim 5, wherein a bottom of the storage body is stepped or one corner of the bottom of the storage body is recessed inward, such that a concave space is formed at the bottom of the storage body; and the temperature control module is detachably arranged in the concave space at the bottom of the storage body, and the temperature control module does not protrude from a radial perimeter surface of the storage body;

wherein:

the battery mounting portion is connected to the refrigeration base, and the battery mounting portion protrudes from the radial perimeter surface of the storage body; or

the battery mounting portion is connected to the radial perimeter surface of the storage body.

7. The temperature-controlled storage device according to claim 5, wherein a bottom of the storage body is stepped or only one corner of the bottom of the storage body is recessed inward, such that a concave space is formed at the bottom of the storage body; the temperature control module is detachably arranged in the concave space at the bottom of the storage body, and at least one side of the temperature control module protrudes from a radial perimeter surface of the storage body;

wherein:

the battery mounting portion is connected to the storage body, and;

the battery mounting portion is located on a side of the storage body away from the temperature control module, or

the battery mounting portion and the temperature control module are located on a same side of the storage body; or

the battery mounting portion is connected to the storage lid.

8. The temperature-controlled storage device according to claim 7, wherein a center of gravity (G) of the temperature control module is away from the storage module, and the center of gravity (G) of the temperature control module is located in a range of 1-1.5 times a radius (R) of a central axis of the storage body.

9. The temperature-controlled storage device according to claim 1, wherein the soft outer package comprises a soft pack body, the soft pack body wraps a storage body of the storage module, and the soft pack body wraps at least a portion of the surface of the temperature control module, and the soft pack body is assembled integrally with the storage body.

10. The temperature-controlled storage device according to claim 9, wherein the soft outer package comprises a soft pack cover, the soft pack cover wraps a storage lid of the storage module, and the soft pack cover is assembled integrally with the storage lid.

11. The temperature-controlled storage device according to claim 9, wherein the soft outer package further comprises at least one ear pouch, wherein:

the at least one ear pouch partially wraps a battery mounting portion of a power supply assembly of the temperature-controlled storage device, or

the at least one ear pouch wraps the battery mounting portion in an openable and closable manner.

12. The temperature-controlled storage device according to claim 10, wherein the storage body and the storage lid define a sealing connection position, the soft pack body and the soft pack cover define an accommodating connection position, and the sealing connection position and the accommodating connection position are arranged in a staggered manner in a height direction; or

the storage body and the storage lid define a sealing connection position, a top of the soft pack body extends upward beyond the sealing connection position, or a bottom of the soft pack cover extends downward beyond the sealing connection position; or

the storage body and the storage lid define a sealing connection position, the soft pack body and the soft pack cover define an accommodating connection position, and the sealing connection position and the accommodating connection position are at a same horizontal level.

13. The temperature-controlled storage device according to claim 5, wherein the storage body is provided with a partition plate, the partition plate divides a storage space of the storage body into a first zone and a second zone, an air outlet for outputting cold air is formed in one of the first zone or the second zone, and the partition plate is rotatably or detachably arranged in the storage body.

14. The temperature-controlled storage device according to claim 5, wherein the refrigeration assembly comprises a compressor, a condenser pipe connected to the compressor, an evaporator pipe connected to the condenser pipe, and a cooling fan arranged on one side of the condenser pipe, and the refrigeration base is provided with at least one pair of ventilation windows for dissipating heat from the refrigeration assembly; wherein

the at least one pair of ventilation windows is arranged on two opposite surfaces of the refrigeration base; or

the at least one pair of ventilation windows is arranged on two adjacent surfaces of the refrigeration base; and

the soft outer package wraps the refrigeration base in a way that avoids ventilation windows of the at least one pair of ventilation windows.

15. The temperature-controlled storage device according to claim 14, wherein the refrigeration base has a mounting surface, the battery mounting portion is arranged on the mounting surface, and the mounting surface is not coplanar with the ventilation windows of the at least one pair of ventilation windows; and

the at least one pair of ventilation windows are arranged on two opposite surfaces of the refrigeration base, and the mounting surface is perpendicular to a plane where the ventilation windows of the at least one pair of ventilation windows are located.

16. The temperature-controlled storage device according to claim 5, wherein the temperature control module further comprises a control circuit, the control circuit is arranged inside the refrigeration base, the battery mounting portion is electrically connected to the control circuit, the control circuit is electrically connected to the refrigeration assembly, an outer side of the refrigeration base is provided with a control panel, the control panel is electrically connected to the control circuit, and the soft outer package wraps the refrigeration base in a way that avoids the control panel.

17. The temperature-controlled storage device according to claim 5, wherein the battery mounting portion has an opening for removing a portable power source, and the opening enables the portable power source to be inserted into the battery mounting portion in a vertically downward direction.

18. The temperature-controlled storage device according to claim 1, further comprising a carrying assembly, wherein the carrying assembly is detachably connected to at least one of the soft outer package or the storage module for a user to carry the temperature-controlled storage device conveniently; and the carrying assembly comprises a human carrying part, and the human carrying part is detachably connected to at least one of the soft outer package or the storage module.

19. The temperature-controlled storage device according to claim 1, further comprising a detachable tabletop, wherein the detachable tabletop is partially supported and unfolded relative to the storage module; or

the temperature-controlled storage device comprises a detachable bracket, and the soft outer package is detachably mounted with the detachably bracket for fixing the soft outer package on a vehicle; or

the temperature-controlled storage device comprises a detachable wheel assembly, wherein the detachable wheel assembly is detachably connected to the soft outer package, and wheels of the detachable wheel assembly are positioned near a bottom of the temperature-controlled storage device for mobility; or

the temperature-controlled storage device comprises a camping expansion assembly, and the camping expansion assembly is disposed on one or more defined by four radial peripheral surfaces of the temperature-controlled storage device for being connected to external accessories.

20. The temperature-controlled storage device according to claim 2, wherein

the power supply assembly comprises one or more portable power sources, and the portable power source is detachably arranged in the battery mounting portion;

the power supply assembly further comprises a photovoltaic power panel and the portable power source electrically connected to the battery mounting portion, and the photovoltaic power panel is configured for supplying power to the portable power source; or

the power supply assembly further comprises at least one

power supply interface configured for being electrically connected to an external power source, such that the external power source supplies power to the temperature control module or the portable power source inside the battery mounting portion, and types of the power supply interface comprise universal serial bus (USB) TYPE-C, alternating current (AC) female plug, delta-shaped interface, and USB interface; and

the temperature control module comprises a control circuit, and the battery mounting portion is electrically connected to the control circuit; when a compressor of the temperature control module is operating, the control circuit controls current inputted to the compressor; and when the compressor of the temperature control mod-

ule is not driven, the control circuit controls current inputted to the portable power source until the portable power source is fully charged.

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